

**Online Reservation, Time Tracking, and Queuing System with AI-Powered
Wait-Time Estimation for One Shot Bar and Billiards**

**A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Ortigas-Cainta**

**In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **Online Reservation, Time Tracking, and
Queuing System with AI-Powered Wait-Time
Estimation for One Shot Bar and Billiards**

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for the degree Bachelor of Science in Information Technology
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APPROVAL SHEET

This capstone project proposal titled **Online Reservation, Time Tracking, and Queuing System with AI-Powered Wait-Time Estimation for One Shot Bar and Billiards**, prepared and submitted by **Zeji S. Arnoco, Karl Andrei S. De Guzman, Michael H. Grace, and Mark Joseph D. Morcozo**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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TABLE OF CONTENTS

	Page
Title Page	i
Endorsement form for Proposal Defense	ii
Approval Sheet	iii
Table of Contents	iv
Introduction	1
Project Context	1
Purpose and Description	4
Objectives	6
Scope and Limitations	8
Review of Related Literature/Studies/Systems	19
Methodology	33
Technical Background	37
Requirements Analysis	48
Requirements Documentation	49
Design of Software, System, Product, and/or Processes	63
References	87
Appendices	91
Transcript of Interview	92
Personal Technical Vitae	104

INTRODUCTION

Project Context

One Shot Bar and Billiards, located in San Juan, Cainta, Rizal, is a cozy recreational spot that has been welcoming customers since its establishment in 2024. Owned by Mr. Raymond Lomabas and supported by a dedicated team of four employees who maintain cleanliness, cater to walk-in customers, and monitor table rentals separated by shifts, the bar has quickly become a frequent destination for students, working adults, and families. With 9 well-maintained billiard tables which are categorized into regular tables and VIP tables with distinct hourly rates — and a warm, inviting atmosphere, One Shot Bar and Billiards offer a relaxing space where customers can unwind, socialize, and enjoy friendly games throughout its 12-hour daily operation from 12 PM to 12 AM. The establishment records a maximum of 108 sessions hourly rented per day reflecting its growing popularity within the community.

One Shot Bar and Billiards often rely on table reservations and hourly usage to manage customer flow and maximize profitability. Despite the growing demand for structured playtime and fair access to tables, they still rely on traditional methods for time tracking and queue management. This process involves writing customer information, arrival times, and booking details on scratch paper. In line with their reservation policy, if the customer does not respond within 15 minutes after the reservation confirmation request is sent, the reservation will be automatically canceled. However, if the customer responds after the 15-minute period, they will be placed next in the queue. Although these manual processes have proven to be functional and effective in their previous operations, the increasing number of customer walk-ins and reservations, particularly during peak hours operation from 7 PM to 8 PM, highlights the need for a more efficient and modernized approach to enhance overall service performance. The absence of a reliable system for monitoring reservations and table usage often results in disorganized customer flow, limited wait-time transparency, and difficulties in effectively tracking table sessions.

One Shot Bar and Billiards currently manage reservations manually and relies mainly on walk-in customers. Customers typically contact the staff through Facebook page for messages or direct chats to inquire about table availability and reservation schedules. As the customer numbers continue to grow, particularly during peak hours, the reliance on manual processes has led to operational challenges. These key processes are outlined below:

1. **Manual Reservation Process** - Customers make reservations either by walking in or by messaging the establishment's Facebook page. Staff manually record these bookings on paper and rely on simple alarms as reminders, which often results in scheduling conflicts, missed messages, and potential double bookings between online reservations and walk-in customers.
2. **Unstructured Customer Queue Management** - When all 9 tables are occupied, the establishment lacks a structured queuing system for managing incoming walk-in customers. Staff currently record customer names on scratch paper and rely on memory to monitor the order of arrivals. This process compromises queue accuracy, increasing the likelihood of customers being skipped and causing potential disputes regarding table priority.
3. **Unautomated Time and Billing System** - Tables are currently monitored by giving customers a paper slip that indicates their time-in, while staff manually operate continuous timers behind the counter without automated alarms. During busy hours, staff often become too occupied to accurately monitor end-times, leading to discrepancies between customers' watches and the staff's timers. Manually calculating the final bill becomes challenging due to varying pricing rates, such as ₱150 for regular tables, ₱200 for VIP tables, and a special ₱150 discount for loyal VIP customers, as well as the strict 1-hour overtime penalty for late extensions. These factors increase the likelihood of billing errors and potential revenue loss.
4. **Inaccurate Waiting Time Estimation** - When the establishment is at full capacity, staff are unable to provide walk-in customers with accurate waiting times. As a result, they rely on rough estimates—for example, informing customers that the wait will be 40 minutes even when a table may actually become available in 30

minutes, in order to avoid underestimating the delay. This reliance on estimation creates uncertainty and frustration among customers waiting. In some cases, customers choose to leave the queue, which may lead to lost sales opportunities for the establishment.

Given the significant challenges One Shot Bar and Billiards face in handling reservations, queue management, time monitoring, and wait-time estimation, there is a pressing need for a more advanced operational solution. To address these concerns, the development and implementation of an integrated management system is proposed, featuring automated reservation handling, real-time table monitoring, queue management, and an AI-powered wait-time estimation. This project proposal aims to modernize their current processes, reduce errors, improve operational efficiency, and provide seamless experience for both staff and customers.

The rationale of this study is grounded in addressing specific operational gaps within the current system. By automating traditional processes, the proposed system reduces manual administrative workload and ensures accurate tracking of session durations. This results in clear and reliable records of playtime, minimizing disputes, and supporting fair billing practices. In addition, it restores queue integrity and reduces ambiguity in wait times, thereby improving overall operational efficiency.

Purpose and Description

The purpose of this study is to improve how One Shot Bar and Billiards manage table reservations, customer queues, and table rentals by replacing the current paper-based process with a centralized web-based system. The proposed system combines reservation handling, real-time table monitoring, queue management, and an AI-powered wait-time estimation into one platform to make daily operations more organized and easier to manage.

The proposed system allows staff to record both online and walk-in reservations digitally, view table availability through a live dashboard, and receive alerts when a session is about to end or has already exceeded the paid time. This helps ensure that playtime is recorded correctly and that customers are charged fairly. To address the problem of customers leaving due to uncertainty about their waiting time, the system uses AI-powered wait-time estimation. Instead of relying on staff estimates, it calculates more accurate waiting times using real data, such as the number of currently served customers and upcoming bookings or reservations. The system uses the First Come, First Served (FCFS) method for managing walk-in customers. This is supported by the study of Implementation of a Website-Based Sports Facility Reservation System Using FCFS Method (2025), which found that the lack of a clear queuing system can cause scheduling conflicts and customer disputes. By applying FCFS, the system ensures that customers are served based on their arrival time, reducing the chances of customers being skipped. The study also applies the concept of real-time queue monitoring from CLIQUE: A Web-Based Queue Management System with Real-Time Monitoring (2022). This study showed that displaying the queue in real time helps staff track customer order more easily and reduces confusion. In the proposed system, a queue dashboard shows the list of waiting customers in order, allowing staff to assign tables properly. In addition, the study is supported by Digital Innovation and Entrepreneurial Success in the Philippine MSME Sector (2025), which explains that small businesses in the Philippines benefit from using digital systems to improve daily operations. For One Shot Bar and Billiards, using a web-based system helps reduce manual work and makes service more consistent.

Overall, the system helps staff manage reservations, table usage, and queues in one place instead of relying on scratch paper and memory, keeping operations organized during peak hours.

Objectives of the Study

General Objectives

To propose an online reservation, time-tracking, queuing, dynamic billing, and AI-powered wait-time estimation system for One Shot Bar and Billiards to enhance business operations and customer experience.

Equipping One Shot Bar and Billiards with a system that integrates table monitoring, reservation handling, queue management, payment processing, and an AI-powered wait-time estimation enables efficient tracking of table usage and organized customer flow. The system provides real-time table availability, automated fractional billing, centralized data management, and data-driven wait-time transparency. This improves operational control, service quality, and overall business efficiency through a unified web-based platform.

Specific Objectives

- 1. To propose an online reservation module that centralizes advance bookings and cross-references live table availability to prevent scheduling conflicts and double bookings.**

The proposed module will provide customers with a web-based interface to book billiard tables in advance. To prevent invalid bookings, the system will require and validate a GCash downpayment before confirming a reservation. Each incoming booking request will be checked against the live availability of the 9 billiard tables to prevent scheduling conflicts and ensure that reserved tables are not mistakenly assigned to walk-in customers.

- 2. To propose a First-Come, First-Served (FCFS) digital queuing module that allows staff to seamlessly register walk-in customers, manage the waiting sequence, and systematically transition them to available tables.**

The proposed module will replace the existing scratch-paper waitlist with a

structured digital queue. Staff will log walk-in customer details, and the system will arrange them by arrival time in FCFS order. A live queue dashboard will allow staff to maintain an accurate waiting sequence. Once a table becomes available, the system will assist staff in assigning the next customer in line to an active session, reducing the likelihood of skipped customers or priority disputes.

3. **To propose a real-time table monitoring and dynamic billing module, deployed on an offline-capable desktop architecture, that provides automated countdown timers and precise payment computations while ensuring continuous operations during unexpected internet interruptions.**

The proposed system will assign an automated timer to each table, accurately calculate additional charges, and minimize errors caused by manual processing. To ensure continuous operation during internet disconnections, the module will be developed as an Electron-based desktop application integrated with local database storage. This setup allows staff to monitor active timers, accommodate walk-in customers, and calculate final bills locally even during outages, while automatically synchronizing data with the cloud once the connection is restored.

4. **To propose an AI-powered wait-time estimation that analyzes active table sessions and upcoming reservations to provide highly accurate wait-time estimates for queued walk-in customers.**

The proposed module will utilize a machine learning algorithm to estimate the waiting time for queued walk-in customers before a table becomes available. Instead of relying on rough staff estimations, the system will analyze historical and real-time session data, such as time of day, party size, and average session length, while also considering upcoming online reservations. This allows the system to generate more accurate and transparent wait-time predictions for walk-in customers, reducing the likelihood of customers leaving before a table becomes available.

Scope and Limitations

This study focuses on the development and implementation of an integrated management system for One Shot Bar and Billiards located in San Juan, Cainta, Rizal. The proposed system aims to automate table reservation processes, including both online advance bookings and walk-in customers. It will also feature automated table timers to monitor game duration, a queue management module to handle waiting customers, and a real-time availability dashboard for staff. The system is intended for use by staff and management to improve operational efficiency and address existing challenges within the establishment.

Scope of the Project

Admin Portal - Web Page

- **Login Page** - Requires administrators to log in using their account credentials to access the admin Portal of the system.
- **Dashboard Page** - Allows administrators to view a complete overview of the entire system.
- **Site Customization Page** - Allows administrators to customize the customer-facing website interface, including announcements. This feature enhances the user experience, ensures consistency with the system's design requirements, and enables administrators to manage and organize announcements more effectively.
 - **“New Announcement” Button** - Allows administrators to add a new announcement by entering its title and content, selecting the announcement type, and setting an expiration date. Administrators may also choose to publish the announcement immediately or save it for later publishing.
 - **Announcement Management Section** - Allows administrators to view all announcements, modify announcement details, change their status to active or inactive, and delete announcements.
- **User Management Page** - Allows administrators to view the total number of users and categorized as admin and managers. Through this page, administrators can efficiently manage user accounts and maintain organized user records.

- **“Add User” Button** - Allows the administrators to create a new user account by entering the staff’s name, username, email address, phone number, and assigned role.
- **User Management Section** - This section allows administrators to view all active and inactive user accounts and manage user information. Administrators are authorized to update user details such as name, username, email address, phone number, and assigned role. They may also reset user passwords to a system-generated default password when necessary. In addition, administrators can archive or restore user accounts.
- **Table Management Page** - Allows administrators to view the total number of tables, active tables, inactive tables, and occupied tables. It provides an overview of table status and includes tools for creating, updating, and organizing table records.
 - **“Add” Button** - Allows the admin to create a new table and assign a name to the table.
 - **Table Management Section** - Allows administrators to view the availability status of each table as available or occupied and provides functionality to update the table name as well as change its status to active or inactive.
- **Promo Codes Page** - Allows administrators to view the total number of promo codes, active codes, used codes, total usage count, and average discount. This helps administrators manage promo codes more effectively by monitoring their usage.
 - **“Generate Code” Button** - Allows administrators to create or generate a promo code, set the discount percentage, add a description, and set the maximum number of uses. They can also set expiry date to the code and choice between activating it immediately after creation, or they can activate it later if they want.
 - **Promo Code Management Section** – Allows administrators to view all generated promo codes, including the number of users who have used each code and the date it was created. The administrator can also copy promo codes and manage their status by activating, deactivating, or deleting them.

- **Rates Editor Page** - Allows administrators to dynamically manage pricing configurations, including hourly rates, happy hour discounts, and overtime charges. The system also enables setting happy hour start and end times, as well as modifying reservation down payment percentages. This functionality helps ensure flexible pricing management and improves operational efficiency.
 - **Rate Preview Section** - Allows administrators to view pricing for one-hour, two-hour, three-hour, and happy hour sessions.
- **Reservation Terms Page** - Allows administrators to configure booking rules, Cancellation policy and terms and conditions. This helps administrators enforce clear booking rules and policies, ensuring consistency, reducing misunderstandings, and improving transparency for users during the reservation process.
 - **Booking Rules Section** - Allows administrators to edit booking rules, including minimum and maximum booking duration, minimum and maximum party size, and cancellation notice period in hours.
 - **Cancellation Policy Section** - Allows administrators to edit the cancellation policy text.
 - **Terms and Conditions Section** - Allows administrators to edit the terms and conditions.
- **Closing Calendar Page** - Allows administrators to view a monthly calendar and mark specific dates as closed. Administrators can specify the reason for each closure and indicate whether it applies to the entire day or only for a selected period. This feature helps manage scheduling and ensures that users are informed of unavailable dates before making a reservation.
 - **Upcoming Closure Management Section** - Allows administrators to view all upcoming scheduled closure dates and delete any entry in case of errors.
- **Analytics Page** - Allows administrators to view table usage analytics. It provides insights into the total revenue, completed session, and total guess served they can filter it by the day, last 7 days, last month, and past year. These analytics help administrators make informed decisions, such as whether to add new tables or optimize existing table availability.

- **“Export CSV” Button** - allows the administrator to export analytics data into a CSV file, displaying metrics and values based on selected filters.
- **Revenue Trajectory Section** - Allows the administrator to monitor revenue trends over time. When the filter is set to Today, it displays revenue on an hourly basis. For the Last 7 Days, revenue is shown per day. When set to Last Month, the system presents weekly revenue data. Lastly, selecting Past Year displays revenue grouped by month, providing a comprehensive view of financial performance over different time periods.
- **Session Distribution Section** - Allows administrators to view the average rental duration for tables, helping them analyze usage patterns.
- **Table Performance Report Section** - Allows administrators to view important data for each table, including the total number of sessions, total revenue, and utilization rate.
- **Activity Log Page** - Allows administrators to monitor overall system activity, including the total number of recorded activities, activities performed today, total reservations made, and total payments processed. It also provides functionality to search for specific activity records and filter results based on date ranges and activity tags for easier tracking.
 - **“Export CSV” Button** - Allows staff to download the entire activity log in spreadsheet format, containing a table with activity log date, type, and description.
- **Customer Feedback Page** - Allows staff to view the total number of messages submitted by customers, including compliments, concerns, suggestions, lost items, and other categories. It also enables staff to search for specific feedback and view detailed information for each entry based on tags. This helps staff efficiently monitor customer feedback, identify issues, and respond appropriately to improve overall service quality.
- **Account Settings Page** - Allows administrators to manage their profile and account information. Administrators can view and update their personal details, including full name, email address, phone number, role, account creation date, username, and password. The system also enables administrators to update their login credentials

to maintain account security.

- **“Staff Portal” Button** - Redirects the administrator to the Staff Portal, where they can access all staff-level features. This allows the administrator to efficiently manage and utilize staff functionalities while simultaneously performing administrative tasks.

Staff Portal - Web Page

- **Login Page** - Requires staff to log in using their account credentials to access the staff Portal of the system.
- **“Live Table Monitor” Button** - Allows staff to monitor the status of all billiard tables within the establishment.
 - **Real-Time Table Dashboard Section** - Provides real-time status updates for each table, indicating whether it is available, occupied, or reserved, and includes an automated timer that begins upon the start of session.
 - **Green** - Available
 - **Orange** - In Use
 - **Blue** - Reserved
 - **Red** - Overtime
 - **Walk-in Queue Section** - This feature displays a real-time list of walk-in customers who are currently waiting for available tables, alongside their AI-powered wait-time estimation.
- **Overview Page** - Allows staff to view the overall table status of the establishment, including table availability such as whether tables are available or occupied, the number of customers in the waiting queue, and the establishment’s total revenue for the day. This feature helps staff monitor operations efficiently.
 - **Table Status Section** - Allows staff to view the status of all billiards tables, indicating whether each table is available, occupied, or reserved.
 - **Upcoming Reservations Section** - Allows staff to view all reservations, including pending and confirmed bookings.
 - **Walk-in Queue Section** - Allows staff to view a real-time list of walk-in customers waiting for available billiards tables, helping staff manage

customer flow efficiently.

- **Table Monitor Page** - Allows staff to monitor and view the status of tables, including available, occupied, and reserved.
 - **“Start Session” Button** - Allows staff to start a session based on the customer’s arrival time or scheduled reservation.
 - **“Extend” Button** - Allows staff to apply an overtime charge policy when a customer exceeds their paid time. If a customer stays beyond their allotted time by 10–15 minutes without requesting an extension, the system automatically classifies it as one half hour of overtime and applies the corresponding charge.
 - **“End” Button** - Allows staff to terminate a customer’s session. It also enables staff to review and remove any automatically applied overtime charges that resulted from staff delays in ending the session, rather than from the customer exceeding the allotted time, before generating the final receipt.
- **Reservation Page** - Allows staff to efficiently manage table reservations, helping streamline booking processes and improve overall customer service.
 - **“Table Bookings” Button** - Allows staff to efficiently manage both online and walk-in reservations. It displays key information such as the total number of daily bookings, pending confirmations, total revenue, and pending payments.
 - **“New Reservation” Button** - Allows staff to manage walk-in reservations by entering customer details such as name, contact number, email address, preferred reservation date and time, duration, party size, and specific table preference.
 - **Customer Reservation Management Section** - Allows staff to view all reservations made by online customers. Staff can verify GCash payment receipts and update reservation statuses accordingly. The system also manages special situations based on the establishment’s policies. For example, staff may approve an “Early Check-in” if the customer arrives ahead of schedule, and a

table is available. In addition, the system can enforce “Automatic Forfeiture,” where the reserved table may be assigned to walk-in customers if the customer is more than five minutes late without notifying the staff.

- **Queue Management Page** - The allows staff to manage the waiting list using a First-Come, First-Served (FCFS) approach, with AI-powered wait-time estimation, especially during peak hours.
- **Waiting Queue Section** - Allows staff to view the current list of waiting customers along with their queue positions in real time.
 - **“Notification” Button** - The system notifies the customer via a speaker when their table is available and ready for use. Upon notification, a 15-minute grace period begins. If the customer fails to arrive within this period, the reservation will be cancelled, and the table will be offered to the next customer in the queue if it has not yet been confirmed.
- **Available Tables Section** - Allows staff to view all available tables, helping them efficiently manage tables and assist customers.
- **Promo Codes Page** - Allows staff to view active promo codes and their usage counts, enabling them to monitor the performance and effectiveness of each promo code.
 - **Promo Codes Management Section** - Allows staff to view all promo codes, whether they are active or inactive, track how many customers have used each code, and see the date each promo code was created.
- **History Page** - Allows staff to view a table of history records and download the entire reservation history in spreadsheet format.
- **Account Settings Page** - Allows staff to manage their profile and account information. Staff can view and update their personal details, including full name, email address, phone number, role, account creation date, username, and password. The system also enables staff to update their login credentials to maintain account security.

Customer Portal - Web Page

- **Home Page** - The landing page serves as the main entry point for customers, providing an overview of the establishment, including rates, operating hours, and the number of tables.
 - **Customer Accounts and Authentication** - Allows customers to optionally register and log in to their accounts. This enables the system to automatically fill in reservation details and securely track their booking history.
 - **“Login” Button** - Allows customers to access their account by verifying their email and password credentials.
 - **“Forgot Password” Button** - Allows customers to access the password recovery section, where they can request a password reset link by entering their registered email address.
 - **“Register” Button** - Allows customers to access the account registration form where they can input personal details such as their name, email, contact number, password, and an optional referral code.
- **Reservation Page** - A step-by-step booking form that allows customers to select their preferred date and time while providing essential details such as name, email address, contact number, party size, and duration. The system is designed to ensure a fast, efficient, and user-friendly reservation experience.
 - **Live Table Status Section** - Allows customers to view all tables with real-time indicators showing their availability such as available or occupied.
 - **Walk-In Queue Snapshot Section** - Allows customers to view the number of people currently waiting in line and the AI-powered estimated wait time for the next available table, helping them make informed decisions on whether to reserve a table or walk in.
 - **Date & Time Picker Section** - A calendar-based system that allows customers to select their preferred date and time for booking. The system

enforces a minimum reservation requirement of at least one (1) hour in advance; otherwise, the booking will not be accepted.

- **“Proceed to Payment” Button** - Allows customers to enter their GCash reference number as proof of payment and upload a screenshot of their down payment receipt directly within the reservation form. After completing the payment, the customer is provided with a reservation ID, which must be saved as it will be used to track the reservation details.
- **Slot Availability Section** - Allows customers to check available time slots. Each slot is limited to five reservations to ensure availability for walk-in customers.
- **My Bookings Section** - Allows customers to track their reservation details by entering their reservation ID. It also enables them to cancel or reschedule their reservation when necessary. If the customer is already logged into their account, all reservation details can be viewed and managed directly without needing to enter a reservation ID. However, reservations that are less than 8 hours away can no longer be canceled or rescheduled. In such cases, the customer’s GCash down payment will be forfeited.
- **Interactive Table Selection** - Customers can view a digital map of the venue displaying all nine billiard tables. The system shows the real-time availability status of each table, indicating whether it is available or currently occupied. Customers can select their preferred table based on its location within the venue. Alternatively, if they have no specific preference, they may choose the “Any Available Table” option, allowing the system to automatically assign the next available table upon their arrival.
- **Rates Page** - A dynamic section that displays pricing for walk-in customers, online reservations, and happy hour promotions. This page also includes a button that redirects users to the Billiards Reservation Page, where the booking form is located. In addition, it provides view to the Reservation Information Section, which contains essential booking details such as the minimum reservation period, down payment requirements, instructions for settling the remaining balance, cancellation policy, accepted payment methods, walk-in queue policy, and applicable extension

charges.

- **About Us Page** - Provides customers with information about the establishment, including its background, location, services offered, and contact information.
- **Feedback Page** - This section contains a form where customers can enter their name, email address, contact number, feedback type, and message. This feature enables the staff and administrator to receive customer feedback, allowing them to address concerns, improve services, and respond to customer needs and requests effectively.

Limitations

- **Customer Portal Web Dependency** - The system is developed as a responsive web application rather than a native mobile application installed directly on a device. As a result, it has limited access to device-specific mobile features and cannot utilize functionalities such as native background processing and operating system-level push notifications without relying on active browser support.
- **Absence of Hardware or IoT Integration** - The automated table timer currently operates solely at the software level, focusing on tracking session duration and calculating billing. Although the system provides visual indicators of table status, such as overtime usage, it is not integrated with physical hardware components such as microcontrollers like Arduino to control real-world devices such as table lighting. As a result, staff are still required to manually manage physical equipment, including turning off the lighting when tables are unoccupied. This limitation reduces the system's potential for full automation and overall operational efficiency.
- **Supabase Service Dependency** - The system relies on Supabase services to manage its database and backend operations. Therefore, the system's availability and functionality depend on the stability and performance of the Supabase platform. Any maintenance activities, server downtime, or technical issues encountered by Supabase may temporarily affect the system's accessibility and performance. During such service interruptions, users may experience limited access to the system and its features.
- **AI Predictive Accuracy Constraint** - The AI Wait-Time Estimation feature calculates expected waiting times using predefined historical data factors such as time of day, day of the week, active sessions, and upcoming reservations. Since the system relies on predictive algorithms, unexpected human behaviors—such as customers unexpectedly ending an ongoing session or groups requesting extended table usage despite an existing queue—may occasionally result in differences between the AI-estimated wait time and the actual waiting time.

Review of Related Literature/Studies/Systems

This literature review examines studies and systems related to reservation management, queue handling, and digital system adoption in service-based establishments related to our proposal. It covers different approaches and findings from existing research, highlighting how digital solutions address the operational challenges common to recreational and service facilities. By analyzing these sources, this section establishes the relevance of developing a web-based reservation system for One Shot Bar and Billiards.

Foreign Related Literature and Studies

Online Reservation Queueing–Inventory System with Two Distinct Services and Client Types by Kumar et al. (2025)

Kumar et al. (2025) present an integrated system that combines online reservation and queueing mechanisms to manage both reserved and walk-in customers within a single platform. The study focuses on structuring service flow so that customers are handled in a defined order, reducing idle time and minimizing the conflict between pre-reserved slots and on-the-spot arrivals. Their findings show that combining a reservation module with a queue system produces a more organized and fair distribution of service across all customer types.

For One Shot Bar and Billiards, where reserved and walk-in customers both compete for the same limited tables during peak hours, this study directly supports the proposed system's approach of integrating table reservation with a queue management feature. Rather than relying on staff memory to determine who gets served next, the system can track both reserved and waiting customers in one place, ensuring that table allocation follows a defined and fair process.

Developing Web-Based Patient Reservation and Data Management System using Rapid Application Design by Subiyakto et al. (2025)

Subiyakto et al. (2025) documented the development of a web-based reservation system

built to replace manual booking processes in a service establishment. Their study highlights the improvements brought about by digitizing reservations, specifically in data accuracy, accessibility, and processing speed. The system allows customers to submit booking requests through a web interface while giving administrators a centralized view of all reservation records, eliminating the errors tied to handwritten logs.

For One Shot Bar and Billiards, where reservations are currently recorded only through a handwritten note, the transition described in this study closely matches what the proposed system aims to accomplish. Moving reservation records from paper to a digital platform addresses the inaccuracy of manual logging and gives staff a clearer, organized record of bookings without having to rely on physical notes.

Implementation of the First Come First Served Method in a Website-Based Sports Facility Reservation Service System by Rangkuti et al. (2025)

This study, published in the International Journal of Engineering, Science and Information Technology, developed a web-based reservation system for a sports facility using the First Come, First Served (FCFS) method. It identifies manual reservation processes as a primary source of scheduling errors and conflicts, noting that the absence of a defined queuing rule causes disputes between customers over table or court access. The FCFS-based system resolved this by assigning service priority based on the order of arrival, reducing conflicts, and improving the overall scheduling process.

For One Shot Bar and Billiards, this study supports the use of FCFS as the queuing logic for walk-in customers waiting for a table. As confirmed in the interview, the establishment currently has no formal way of tracking who arrived first, which leads to customers being skipped unintentionally. Applying FCFS within the proposed system ensures that waiting customers are served in the order they registered, removing the need for staff to rely on memory when managing the queue.

Online Events Booking and Reservation System by Ganiyu et al. (2024)

This study, published on ResearchGate, presents an online booking platform for events that

replace informal reservation practices with a structured digital system. It covers customer data management, real-time availability tracking, and automated booking confirmations, all of which address the delays and disorganization tied to manual booking methods. The platform allows administrators to view all reservation details in one place and update availability without requiring additional staff effort.

For One Shot Bar and Billiards, the real-time availability tracking and customer data management features described in this study are directly applicable to the proposed system's table availability dashboard. Since the establishment currently has no way of showing customers which tables are open or how many are waiting, a system that displays this information and stores customer details in a database would directly address the organizational gaps identified during the interview.

Development of a Web-Based Reservation System to Improve the Efficiency of Catering Services by Sahara et al. (2025)

This study, published on ResearchGate, developed a web-based reservation platform for catering services with the goal of reducing manual errors and streamlining the booking process. The researchers found that handwritten records were a consistent source of inaccuracies, missed bookings, and delayed confirmations. The digital system replaced these with automated scheduling, customer data storage, and a clear booking workflow that reduced the workload on staff while improving the reliability of reservation records.

For One Shot Bar and Billiards, this study reinforces the case for replacing the establishment's current paper-based reservation method. The same problems identified in this study — manual errors, unreliable records, and heavy reliance on staff effort — are present at One Shot Bar, as confirmed in the interview. A web-based system that handles customer details and reservation data digitally would directly resolve these issues.

Queue Wait Time Prediction in High Performance Computing (HPC) Systems by Okafor et al. (2025)

Okafor et al. (2025), published in The Journal of Supercomputing (Springer Nature),

investigate the use of supervised machine learning algorithms to predict job wait times in leadership-class High Performance Computing systems. The study analyzes over 900,000 job log samples spanning several years and examines how job characteristics — including the number of nodes requested, wall time, queue assignment, and other resource requirements — influence how long a submitted job waits before execution begins.

For For One Shot Bar and Billiards, this study provides a strong technical precedent for the core principle behind the proposed AI wait-time estimation module: that accurate prediction of how long a resource-constrained queue will take depends on the characteristics of the jobs — or in this case, the sessions — currently in that queue. Just as HPC systems use job size, resource type, and queue position to estimate wait time, the proposed system uses session duration patterns, table occupancy, party size, time of day, and upcoming reservations to predict when a walk-in customer will be assigned a table.

Waiting Time Prediction in Queue Management: Leveraging Machine Learning Approach by Taton et al. (2024)

Taton et al. (2024), presented at the International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET) and published on IEEE Xplore, evaluate multiple machine learning models for predicting customer wait times in queue management systems. The study compares regression algorithms including Long Short-Term Memory (LSTM) networks and the Voting Regressor, with the latter achieving an R^2 score of 0.944 and a mean absolute error of only 1.828 minutes — the best performance among all models tested. The researchers demonstrate that AI-based queue management systems can dynamically allocate customer requests to available service counters, significantly optimizing resource utilization compared to traditional manual methods.

For One Shot Bar and Billiards, this study directly validates the use of machine learning for real-time wait-time estimation in service queue environments. The same data inputs that drove predictions in this study — such as queue length, arrival rate, and service rate — mirror the data the proposed system collects through active table sessions and walk-in customer registrations. The findings confirm that machine learning can produce accurate,

real-time wait estimates that outperform staff-based guessing, which is the precise problem identified at One Shot during peak hours.

Turf Booking System: Simplifying Sports Facility Reservations by Chatterjee et al. (2025)

Published in the International Journal for Multidisciplinary Research (IJFMR), this article presents a web-based booking system for a turf-based sports facility. The researchers address three specific operational problems: double bookings from overlapping manual entries, scheduling conflicts caused by the absence of a live availability check, and the lack of real-time monitoring tools for administrators. The system resolved these by introducing an automated reservation platform with a live availability dashboard that staff could use to track facility usage.

For One Shot Bar and Billiards, the problems described in this study are consistent with what was identified during the interview. The establishment has no way of preventing double bookings or displaying table availability to wait for customers. The table availability dashboard and reservation management features proposed for One Shot directly address the same operational gaps this study documents.

Design and Implementation of Computerized Restaurant Table Booking System by Egigogo et al. (2024)

Published in the Ceddi Journal of Information System and Technology (JST), this study presents the development of a computerized table booking system for a service establishment that previously relied on walk-in and verbal reservations. The researchers demonstrate how digitizing the table booking process reduced manual errors, shortened processing time, and gave staff a structured workflow for handling reservations. The system replaced informal booking practices with a defined digital process that both staff and customers could follow.

For One Shot Bar and Billiards, this study provides a direct comparison. The establishment currently accepts reservations verbally and records them on paper, which is the same

starting condition described here. The demonstrated reduction in booking errors and processing time in this study supports the rationale for the proposed billiard table reservation system, which aims to replace this informal process with a structured digital workflow.

Pura Kencana Sports Centre Management System by Jafridin and Salamat (2023)

Published in Applied Information Technology and Computer Science (AITCS), Volume 4, Number 1, this article presents the development of a web-based court booking and management system for Pura Kencana Sports Centre in Malaysia, which had previously relied on a paper-based system. The researchers identified three specific problems: duplicate bookings due to the lack of a centralized record, the inability of customers to check court availability without contacting staff, and the frequent loss of customer information under the manual setup. The system resolved these by implementing a web-based platform under an Agile methodology, with a focus on court booking management and overall system performance evaluation.

For One Shot Bar and Billiards, the operational problems documented in this study are consistent with what was observed in the establishment. Customers have no way to verify table availability without asking staff directly, and the absence of a centralized booking record exposes the establishment to the same duplicate booking risk this study describes. The court availability tracking and centralized management panel developed for Pura Kencana directly corresponds to the table availability dashboard and reservation management workflow proposed for One Shot Bar and Billiards.

Local Related Literature and Studies

Appoint Deportes: An Online Reservation System for Sports Facilities by German et al. (2024)

German, Bautista, et al. (2024), presented at ICCMB 2024 and published in the ACM

Digital Library, developed a web-based reservation system specifically for sports facilities in the Philippines. The study addresses the lack of accessible digital platforms for booking recreational and training spaces, noting that most local sports facilities still rely on in-person or informal reservation methods. The system provides customers with a structured online booking process and gives administrators a centralized dashboard for managing facility availability.

For One Shot Bar and Billiards, this study is the most directly comparable for local reference. Both are Filipino recreational establishments that rely on manual reservations. The operational gaps addressed in this study — the absence of a digital booking platform, the difficulty of managing walk-in volume, and the need for organized customer records — match the problems documented at One Shot Bar through the interview.

SORMS: Steffi's Online Reservation Management System by Balintag et al. (2025)

Published in Philippine EJournals, this study documents the development and evaluation of an online reservation management system for a local service establishment. The system was evaluated using ISO 25010 standards and showed a measurable reduction in booking errors and faster processing times compared to the manual method it replaced. The researchers attribute these improvements to the system's ability to organize and store reservation data digitally, removing the inconsistencies caused by handwritten records.

For One Shot Bar and Billiards, the results of SORMS support the practical case for the proposed system. The establishment currently processes reservations by writing names and contact numbers, a method that is slow and prone to gaps. A system evaluated to reduce booking errors and processing time, as demonstrated here, directly addresses the reliability problems present in One Shot Bar's current reservation process.

CLIQUE: A Web-Based Queue Management System with Real-Time Queue Tracking and Notification of Units for Angeles University Foundation Office of the University Registrar by Mallari et al. (2022)

Published by IEOM Society International, this Philippine university study presents a web-

based queuing system designed to address the inefficiencies of manual queue management in service environments. The system uses real-time monitoring to track queue position and service status, ensuring that customers are attended in a fair and organized order. The researchers found that real-time queue visibility reduced disputes, improved customer flow, and allowed staff to focus on service rather than manually tracking who was next.

For One Shot Bar and Billiards, this study directly mirrors the queue management challenges identified during the interview. The establishment loses track of customer arrival order during peak hours, leading to customers being skipped and staff struggling to manage who is next. The real-time queue monitoring approach presented in CLIQUE supports the development of the proposed system's queue dashboard, which aims to give staff a clear, live view of waiting customers and their order.

Booking and Reservation System: A Unified Application Using Location-Based Services for Sustainable Tourism Networks by Gosela and Encarnation (2024)

Published on ResearchGate, this Philippine study presents a unified booking and reservation application developed to replace informal manual booking practices in local establishments. The system features real-time availability tracking and a structured reservation workflow, both designed to give customers and administrators a reliable way to manage bookings without relying on paper records or verbal confirmations. The researchers found that digitizing the booking process reduced errors and gave establishments better control over their availability data.

For One Shot Bar and Billiards, the reservation logic and real-time availability features in this study closely match the proposed system's billiard table reservation and availability monitoring functions. Both address the same core problem: a local Filipino business using informal booking methods that produce unreliable records. The approach demonstrated here supports the feasibility and direction of the proposed system.

Digital Innovation and Entrepreneurial Success in the Philippine MSME Sector: Challenges and Opportunities by Cordova et al. (2025)

Published on ResearchGate, this Philippine study examines the relationship between digital innovation and business performance among micro, small, and medium enterprises (MSMEs). The researchers identify the shift from manual to digital operations as a critical factor in improving efficiency and competitiveness, noting that small Filipino businesses that adopt digital tools are better positioned to manage their operations and serve customers consistently. The study also addresses the challenges MSMEs face in transitioning to digital systems and what factors support successful adoption.

For One Shot Bar and Billiards, a locally owned small establishment, this study provides a broader context for why the proposed system is a relevant and timely development. The shift from manual reservation handling to a web-based platform is precisely the kind of digital transition this research identifies as beneficial for Philippine MSMEs. The study supports the long-term value of the proposed system beyond its immediate operational improvements.

The SME investment paradox: How to grow while cutting costs in 2026 by Carandang (2026)

This feature article from the Philippine Daily Inquirer discusses the shift toward data-driven operations among Filipino MSMEs in 2026. It points out that digitizing records alone is no longer sufficient; small businesses operating recreational spaces are now expected to track table occupancy trends, identify peak-hour bottlenecks, and use usage data to support operational decisions. The article frames automated monitoring as a practical tool for small business owners who need to make decisions based on actual patterns rather than estimates.

For One Shot Bar and Billiards, this article supports the system's goal of generating usage data through automated table monitoring. The establishment currently has no way of tracking how long tables are occupied or when peak congestion typically occurs. A system that records this data over time gives management a factual basis for operational decisions, such as staffing adjustments during peak periods.

How Queue Management Systems Reduce Waiting Time for Philippine Businesses by

QueueBee Philippines (2026)

QueueBee Philippines (2026), published on the QueueBee Philippines Solutions blog, examines how digital queue management systems address the persistent challenge of long and unmanaged waiting times faced by Philippine businesses across industries including banking, healthcare, retail, and public services. The article outlines how a QMS streamlines customer flow by organizing service orders, providing real-time queue status updates, and offering estimated wait times calculated directly from live queue data. It further identifies unmanaged queues as a direct cause of customer walkouts and revenue loss, and positions smart QMS deployment as both an operational and customer satisfaction solution.

For One Shot Bar and Billiards, this article provides direct, locally grounded support for the proposed system's queue management and wait-time estimation features. The operational problems it describes — customers leaving due to uncertain wait times and staff overwhelmed during peak hours — precisely mirror the conditions observed at One Shot during its busiest hours. The article's emphasis on providing estimated wait times calculated from current queue data is the exact mechanism the proposed AI wait-time estimation module aims to automate and refine walk-in customers waiting for an available table.

Unsolvable Billiards Problem? by Velasco (2026)

This sports feature from The Philippine Star discusses the gap between the talent level of Filipino billiard players and the management infrastructure of local billiard venues. The article points out that local halls generally lack the scheduling, timing, and monitoring systems found in international venues, which creates friction in day-to-day operations. It frames the absence of digital management tools as a structural problem that affects both the player's experience and the operational performance of local billiard establishments.

For One Shot Bar and Billiards, this article provides direct industry context for the proposed system. The operational problems it describes — lack of scheduling tools, no timing systems, and poor monitoring — are the same issues identified during the interview with the establishment's management. The article supports the argument that modernizing

a billiard hall's management system is not an isolated improvement but part of a broader need across local venues in the Philippines.

When Authenticity Meets the Machine by Jabal (2026)

This article from Philstar.com examines how small Filipino businesses are using automated systems to reduce their reliance on manual judgment in day-to-day operations. It argues that tools like queue alerts and automated timers allow staff to focus on direct customer service instead of managing administrative tasks by hand. The article describes this as a practical balance between automation and human service, where the system handles tracking while staff handle interaction.

For One Shot Bar and Billiards, this article supports the proposed system's timer and queue management features. As raised in the interview, staff currently have difficulty monitoring how long each table has been in use, which leads to disputes over billing and overstaying customers. Automating this through a table timer feature addresses the same problem the article describes: staff spending effort on tracking when a system could handle it instead.

How technology will supercharge the Philippines' MSMEs by Tanri (2026)

This report from The Manila Times focuses on how integrated digital systems — where reservations, payments, and usage data are connected on a single platform — give small business owners a real-time picture of their operations. It highlights that linked systems allow owners to review performance data and make decisions based on current information rather than end-of-day estimates. The report frames this type of integration as a competitive necessity for Philippine MSMEs in 2026.

For One Shot Bar and Billiards, this article directly supports the case for building an integrated management system rather than addressing only one operational problem in isolation. The proposed system connects table reservations, queue tracking, and usage monitoring in one platform, which mirrors the kind of integrated tool the article describes as valuable for small business operations. This supports the broader scope of the proposed system as a unified management solution.

Synthesis of the Literature Review

The studies and articles reviewed in this section share a consistent finding: manual processes in service-based establishments produce the same category of problems regardless of the setting. Missed bookings, unclear queue order, inaccurate time records, and the absence of organized availability data are documented across billiard halls, sports facilities, catering services, restaurants, and public-facing offices in both foreign and local contexts. This pattern is directly applicable to One Shot Bar and Billiards, where the interview with Mr. Aya-ay confirms the same conditions — reservations recorded through chat and paper notes, queue order managed by memory, table time tracked through physical slips, and no platform for customers to check table availability before arriving.

On the reservation side, Subiyakto et al. (2025), Sahara et al. (2025), and Egigogo et al. (2024) each demonstrate that replacing handwritten booking records with a web-based system reduces errors and removes the dependency on staff recall. Jafridin and Salamat (2023) and Chatterjee et al. (2025) extend this to facility-specific contexts, showing that sports venues with paper-based booking systems face duplicate entries and scheduling conflicts that a centralized digital platform resolves. Ganiyu et al. (2024) adds real-time availability tracking and automated booking confirmations as features that directly reduce the disorganization tied to informal reservation methods. These findings together support the reservation module proposed for One Shot, which centralizes booking records and checks each incoming request against live table availability to prevent double bookings.

For queue management, Rangkuti et al. (2025) establishes First Come, First Served as the appropriate queuing logic for walk-in customers in a facility reservation context, noting that the absence of a defined rule causes disputes. Kumar et al. (2025) reinforces this by showing that a single platform can manage both reserved and walk-in customers in a structured order, reducing idle time and service conflicts. Mallari et al. (2022) demonstrates in a local Philippine context that real-time queue visibility allows staff to manage customer flow without relying on memory. QueueBee Philippines (2026) further supports this with direct evidence from the Philippine business environment, identifying unmanaged queues as a cause of customer walkouts and framing estimated wait times drawn from live queue

data as a practical operational tool. These studies collectively support the proposed system's queue dashboard, which registers walk-in customers in numbered order and displays their position and wait status to staff.

On wait-time estimation, Taton et al. (2024) evaluated multiple machine learning models for predicting customer wait times in queue environments and found that AI-based systems significantly outperform manual staff estimates. Okafor et al. (2025) provides further technical grounding, showing that accurate wait-time prediction depends on the current characteristics of items already in the queue — the same principle the proposed AI estimation module applies using active session data, table occupancy, party size, and upcoming reservations. Jabal (2026) adds local context, noting that automated queue alerts and timers allow staff to redirect their attention to direct customer service rather than administrative tracking. This is consistent with the problem raised in the interview, where staff currently estimate wait times manually based on a running physical timer.

For table monitoring and time tracking, the interview confirms that One Shot currently distributes paper slips to customers to track session start time, with no automated alert when time runs out. Velasco (2026) frames this as a structural gap common to local billiard venues, noting that the absence of scheduling and timing systems creates friction in operations. Jabal (2026) connects this directly to the value of automation — when a system handles time monitoring, staff can focus on service rather than watching a timer. The proposed system's table session module addresses this by recording session start times digitally and tracking usage without requiring a separate physical record.

On the broader question of digital adoption for small Filipino businesses, Cordova et al. (2025), Carandang (2026), and Tanri (2026) each address the value of integrated digital systems for Philippine MSMEs. Cordova et al. (2025) identifies the shift to digital tools as a key factor in improving operational consistency for small businesses. Carandang (2026) pushes further, pointing out that local businesses operating recreational spaces now benefit from tracking usage patterns and peak-hour data rather than relying on daily estimates. Tanri (2026) frames integrated platforms — where reservations, usage, and operational data are connected — as a practical necessity rather than an optional upgrade. One Shot

Bar and Billiards, as a locally owned establishment without any digital management tools, fits directly within the context these articles describe.

Among local studies, Appoint Deportes (German et al., 2024) and Gosela and Encarnation (2024) both document Filipino recreational establishments transitioning from informal booking methods to structured digital platforms, addressing the same operational gaps observed at One Shot. SORMS (Balintag et al., 2025) provides evaluation-level evidence that these transitions measurably reduce booking errors and processing time. Taken together, the local literature confirms that the proposed system addresses a recognized and recurring gap in how Filipino small businesses currently handle service management, and that digital reservation and queue platforms have already demonstrated practical results in comparable local settings.

METHODOLOGY

Agile Methodology Using Kanban Framework

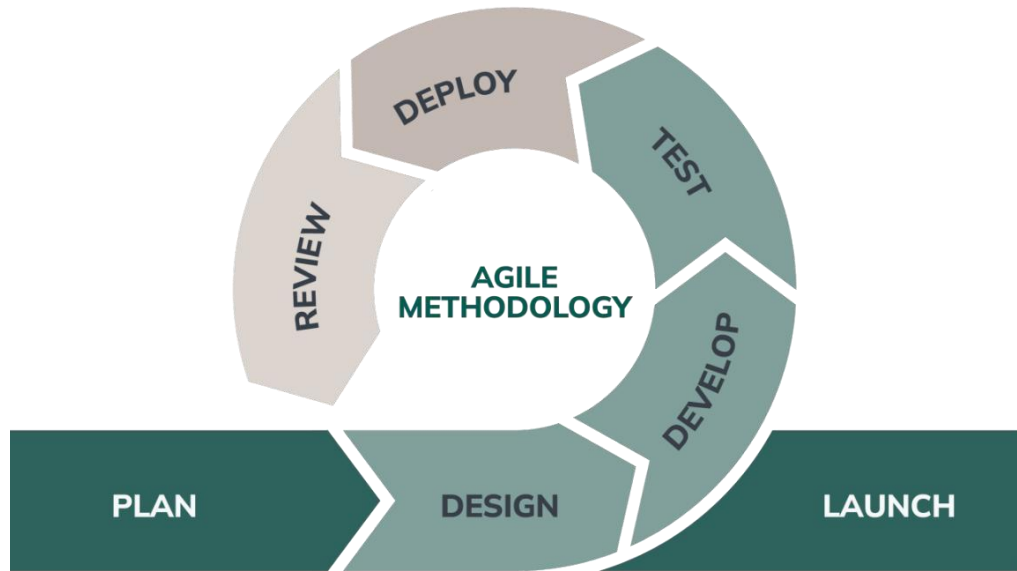


Figure 1. *Agile Methodology*

The development of the Online Reservation, Time Tracking, and Queuing System with AI-Powered Wait-Time Estimation for One Shot Bar and Billiards adopts the Agile methodology Using Kanban framework. Agile is an iterative and incremental approach to software development that prioritizes continuous delivery, collaboration, and adaptability to change. Rather than completing all phases in a fixed sequence before moving forward, Agile divides the project into manageable cycles that allow the team to build, evaluate, and improve the system progressively throughout the development period.

This methodology is chosen because it allows the proponents to respond quickly to changes in requirements, improve features incrementally, and continuously align the system with the operational needs of One Shot Bar and Billiards. By visualizing tasks and managing workflow efficiently, the Kanban framework ensures that development remains organized, transparent, and flexibility.

1. **Plan:** The proponents initiated the project by analyzing the existing manual operations of One Shot Bar and Billiards through direct interviews with the management. During this phase, the key operational challenges were identified, and the system requirements such as the integration of an AI-based wait-time estimation tool and offline system functionality during internet interruptions were carefully planned and defined.
2. **Design:** The System Design phase was prepared by gathering information and analyzing the required specifications identified during the planning phase. The proponents designed the system architecture, developed the Entity-Relationship Diagram (ERD), and created visual prototypes of the user interface. This phase also included designing the interactive floor plan for table selection and mapping the distinct user experiences for both the Customer Web Portal and the Staff Desktop Application.
3. **Develop:** The system will be developed module by module while carefully considering the necessary specifications for future reference. The proponents actively coded the system by programming the online reservation workflow, implementing real-time table timers, integrating an AI API for queue prediction, and packaging the staff portal into a standalone executable application. Each module is integrated iteratively to significantly enhance the overall system development and functionality.
4. **Test:** Each module developed from the implementation phase will be tested and integrated into the system. The proponents conducted rigorous testing on specific functionalities, such as verifying GCash reference numbers and ensuring the local database successfully synchronized data back to the cloud after simulated offline testing. Post-integration is checked for any failures, glitches, bugs, or faults before deployment.
5. **Deploy:** Once a stable version of the system successfully passed testing, it proceeded to the deployment phase. The Customer Portal was hosted on a live cloud server to enable public web access, while the Admin and Staff Portal was installed as a local desktop application on the establishment's hardware.
6. **Review:** The proponents presented the deployed system to the management of One Shot Bar and Billiards to gather operational feedback. This phase allowed the team to evaluate whether the deployed modules effectively addressed the initial problems, such as queue integrity and billing accuracy, and to make necessary adjustments based on the staff's direct experience.

7. **Launch:** This phase thoroughly evaluates the System to ensure that it is fully functional and that the objectives of the proponents are achieved through optimal performance in a live environment. Once finalized and refined, the system is officially turned over to the establishment for day-to-day operations. Continuous monitoring is conducted to identify errors and implement patches or hotfixes, ensuring the system's reliability, maintainability, and extended lifespan.

Agile Using Kanban Process

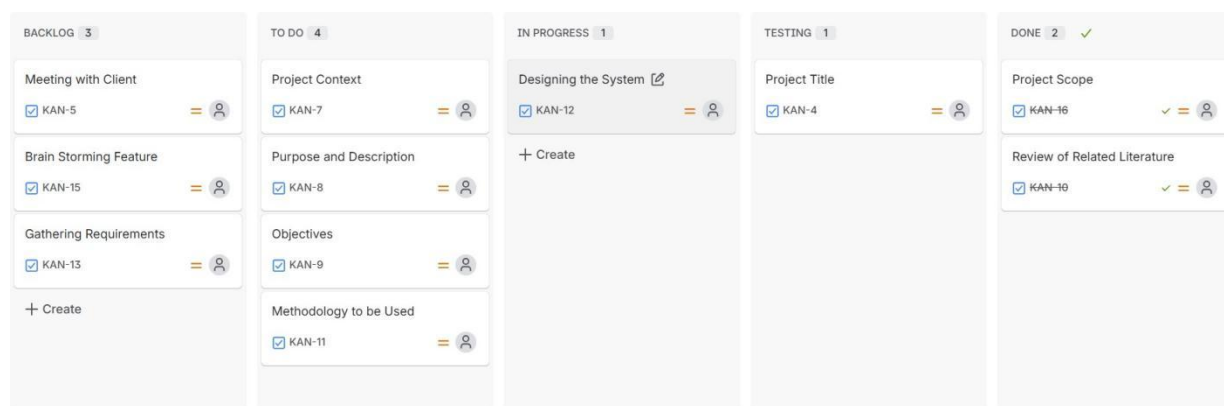


Figure 2. *Agile Methodology with Kanban Boards*

A Kanban board was utilized to visualize and manage the development workflow, enabling the proponents to monitor the progress of each task in real time. Tasks were categorized into different stages according to their development status, ensuring an organized and systematic approach to the system development process.

1. **Backlog:** The Backlog stage serves as the main source of all identified tasks and requirements. This includes activities such as creating the project title, meeting with the client, brainstorming features, and gathering requirements. This stage helps the proponents establish a clear direction on what needs to be done, ensuring a smoother and more organized development process.
2. **To Do:** The To Do stage serves as the phase where tasks are prepared and ready to be initiated. This includes activities such as defining the project context, writing the purpose

and description, setting objectives, conducting the Review of Related Literature (RRL), and determining the methodology to be used. This stage ensures that all necessary components are organized, providing a structured and well-documented foundation. It prepares everything needed before the actual development begins.

3. **In Progress:** The In Progress stage is where the proponents actively develop the system. This phase involves designing and implementing the user interface (UI) and defining the user experience (UX). It also includes refining wireframes, developing system features, and improving documentation. The team follows a work-in-progress (WIP) approach to avoid overloading and ensures that only a manageable number of tasks are handled at a time to maintain efficiency.
4. **Testing:** The testing stage is one of the most important phases of system development before deployment. It involves verifying whether the completed tasks have been approved by the adviser, identifying and resolving errors or bugs within the system, and evaluating whether the developed features meet the specified requirements and objectives. This stage also includes conducting functionality testing, usability testing, and system validation to ensure that the web system operates properly, efficiently, and reliably. Feedback gathered during this phase is used to improve the overall quality and performance of the system before proceeding to the final stage.
5. **Done:** The Done stage is the final stage of the developed system, where all tasks are completed, tested, and approved. It includes finalized outputs such as the project scope, review of related literature (RRL), and the developed web system. Only components that have passed the testing stage and meet all requirements are considered done. At this stage, all outputs are finished and ready for deployment and use in real operations.

The Agile methodology using the Kanban framework provides a structured yet flexible approach to system development. Since the system involves several modules that need to be developed alongside ongoing documentation requirements, a sequential methodology would be inefficient and prone to delays. Kanban allows the team to respond to feedback immediately — whether from consultations with the adviser, or testing findings — without having to restart a formal phase.

Technical Background

Technologies to be Used

The development of One Shot Bar and Billiards system integrates modern technologies to provide an efficient, reliable, and user-friendly platform for both customers and staff. By utilizing a combination of web development tools, programming languages, database systems, and supporting technologies, the proposed system ensures real-time operation, reliable data handling, and smooth user interaction. The selected technology stack prioritizes usability, scalability, security, and performance to meet the operational needs of the establishment. The following technologies are used in the development and implementation of the system.

Resources

Hardware

- **Laptop:** A laptop is a portable, general-purpose computing device designed to process, store, and manage data efficiently. It integrates essential components, including a display, keyboard, processing unit, and storage, into a single compact system. Laptops are widely used in various applications such as document creation, software development, data analysis, and multimedia processing.
- **Mobile Device:** A mobile phone is a handheld electronic communication device that supports voice calls, text messaging, and internet connectivity. Modern smartphones extend these functionalities by incorporating features such as high-resolution cameras, mobile applications, and high-speed data access. As a result, they function as compact computing devices capable of performing a wide range of tasks traditionally associated with personal computers.

Laptop Specifications	
Operating System	Windows 11 Home
Processor	Intel Core i5 12th Generation
RAM	8 GB
Storage	512 GB
Display	1920x1080p pixels, 1080p 144Hz
Graphics	GTX 1650
Internet Speed Connection	500 Mbps

Table 1. *Hardware Specifications used for Software Development*

Minimum and Recommended Specifications for the System

Mobile Phones / Tablets		
Component	Minimum Specification	Recommended Specification
Operating System	Android 7.0+; iOS 12.4+	Android 10+; iOS 14+
Processor	Quad-core 1.4 GHz	Octa-core 2.0 GHz
RAM	2 GB	4 GB
Storage	32 GB	64 GB
Screen Resolution	1280x720	1920x1080
Internet Speed Connection	Wi-Fi or mobile cellular data capable	Wi-Fi or mobile cellular data capable

Table 2. *Minimum and Recommended Specifications for Mobile Devices*

Laptops / Personal Computers (PCs)		
Component	Minimum Specification	Recommended Specification
Operating System	Windows 7 SP1 or later; macOS 10.9 or later	Windows 10 or later; macOS 10.14 or later
Processor	1 GHz or faster with 2 or more cores (64-bit)	Intel Core i5 or equivalent
RAM	4 GB	8 GB
Storage	128 GB	256 GB
Screen Resolution	1024x768 pixels	1920x1080
Internet Speed Connection	Broadband connection (Cable, DSL, or faster)	Broadband connection (Cable, DSL, or faster)

Table 3. *Minimum and Recommended Specifications for Computing Devices*

General Software Tools

- **Operating System - Windows 10:** Developed by Microsoft and released in July 2015, Windows 10 is a widely used operating system known for its user-friendly interface. It offers features such as the Start Menu, virtual desktops, and multitasking capabilities, which facilitate efficient workflow and software develop
- **Operating System - Windows 11:** Launched by Microsoft in October 2021, Windows 11 is the successor to Windows 10. It introduces a modernized interface with rounded corners, a centered taskbar, enhanced themes, and improved performance features, providing an optimized environment for both development and testing.
- **Internet Browser:** An internet browser is a software application that enables users to access and view content on the World Wide Web, including websites, images, videos, and other online resources. In this project, it was used to run, test, and verify the functionality of the web application.

Programming, Scripting, and Markup Languages

- **TypeScript:** TypeScript is the primary programming language used throughout the system. As a statically typed superset of JavaScript, it enables the proponents to catch errors early during development, enforce data types across components, and maintain a more organized and scalable codebase. All major source files in the system are written in TypeScript using the TSX syntax, which combines TypeScript with JSX for building React components.
- **HTML:** HTML serves as the foundation for organizing and structuring the system. It ensures that essential features such as booking forms, user interfaces, and customer information are clearly presented and easy to navigate. Through proper structuring, HTML helps maintain a logical flow of content across all pages of the system.
- **CSS:** The proponents use CSS to enhance the visual design and layout of the system. It ensures a responsive and user-friendly interface by allowing the pages to adapt smoothly to different screen sizes, whether on desktop or mobile devices.
- **JavaScript:** JavaScript is used in supporting configuration files and service worker scripts, such as the One Signal push notification worker. It serves as the runtime foundation upon which TypeScript is compiled and executed in the browser.
- **Node.js:** A cross-platform runtime environment that executes JavaScript code outside a web browser. It serves as the foundational development environment for the system, allowing the proponents to run a local server, manage dependencies via npm (Node Package Manager), and compile both the React web application and the Electron desktop application.

Libraries, Frameworks, and Databases

- **Supabase:** An open-source backend-as-a-service that provides a PostgreSQL database, authentication, real-time data handling, and API generation. It is used to manage application data, handle user access, and support real-time updates within the system.
- **Tailwind:** A utility-first CSS framework that provides predefined classes for styling components directly in the markup. It is used to apply consistent spacing, typography,

colors, and responsive layouts throughout the system without writing custom CSS from scratch.

- **React:** A JavaScript library for building user interfaces using a component-based architecture. React serves as the core frontend framework of the system, enabling the proponents to create reusable UI components and manage dynamic content efficiently across all pages.
- **JSON:** JSON is used for project configuration and dependency management files, particularly in `package.json` and `package-lock.json`. It defines the project's metadata, installed libraries, version constraints, and builds scripts that govern how the system is set up and running.
- **React Router:** A routing library for React applications that enable customer-side navigation without full page reloads. It is used to manage the different pages and views of the system, including role-based layouts for administrators.
- **React Hook Form:** A lightweight library for managing form state and validation in React. It is used to handle user inputs across booking forms, settings of modals, and other interactive form elements within the system.
- **Material UI (MUI):** A comprehensive React component library that follows Google's Material Design guidelines. It provides ready-made components such as icons, buttons, and layout elements that are used to supplement the system's interface.
- **Motion (Framer Motion):** An animation library for React that enables smooth transitions and motion effects. It is used to add visual feedback and fluid animations to interactive elements within the system.
- **Recharts:** A charting library built on top of React and SVG. It is used to render data visualizations such as graphs and charts, particularly within the administrative dashboard of the system.
- **Sonner:** A toast notification library for React that displays brief, non-intrusive messages to users. It is used to provide real-time feedback on actions such as successful bookings, errors, or system updates.
- **React OneSignal:** A React integration for the OneSignal push notification service. It is used to send and manage a browser to push notifications to users of the system.

- **EmailJS:** A client-side email service library that allows the system to send emails directly from the browser without requiring a dedicated mail server. It is used for sending notifications or confirmation messages to users via email.
- **Date-fns & React Day Picker:** A date utility library paired with a calendar picker component. Together, they are used to handle date formatting, calculation, and selection within booking and scheduling features of the system.
- **Canvas Confetti:** A lightweight JavaScript library for rendering confetti particle animations. It is used to add celebratory visual effects triggered by specific user interactions within the system.
- **Vite:** A modern, high-performance frontend build tool. The proponents utilized Vite to initialize and bundle the React and TypeScript source code, providing significantly faster loading times and hot-module replacement (HMR) during the development phase compared to traditional bundlers.
- **Electron.js:** A framework used to build cross-platform desktop applications using web technologies like JavaScript, HTML, and CSS. The proponents utilized this tool to package the Admin and Staff web portals into a standalone executable desktop application (.exe), allowing it to run smoothly on the establishment's local hardware outside of a standard web browser.
- **IndexedDB:** A low-level API for client-side storage of significant amounts of structured data. This was utilized to enable the offline capabilities of the staff application. It temporarily stores walk-in queue data, active table timers, and billing records locally when the internet disconnects, ensuring operations continue without data loss until the cloud database can be synchronized.

DevOps Tools

- **GitHub:** GitHub was selected as the platform for repository hosting and collaboration. It enables developers to store code, manage branches, and review changes through pull requests. Additionally, GitHub supports team collaboration with features like issue tracking and project management, helping maintain an organized and efficient development workflow.

Design and Prototyping Tools

- **Figma:** The proponents utilized Figma as the primary tool for designing and prototyping the system's user interface. It enabled the team to collaboratively create, refine, and test interface designs in real time, ensuring consistency and a user-friendly experience throughout the application.
- **Canva:** Canva was used for creating digital assets and presentation materials. Its accessibility and wide range of design templates allowed the team to produce professional and visually engaging outputs to effectively communicate the system's features and overall concept.
- **Draw.io:** Draw.io was utilized to create the necessary diagrams for the capstone proposal, including use-case diagrams, process flowcharts, and the system architecture representations. It allows the proponents to visually present the system's structure, user interactions, and workflow processes in a clear and organized manner.
- **FlatIcon:** FlatIcon was utilized to provide the icons used in creating the diagrams for the capstone proposal, including system architecture and requirement analysis. It offers a wide range of flat icons that help visually represent users, devices, and system components in a clean and consistent design. Through these icons, the proponents were able to enhance the clarity and presentation of the system's structure, interactions, and workflow processes.
- **Miro:** Miro was utilized for creating flowcharts, conducting brainstorming sessions, and supporting collaborative planning activities related to the capstone project. It enabled the proponents to visually organize ideas, map out workflows, and design the system process flowcharts within a collaborative online environment. Through its interactive workspace and diagramming tools, the team was able to clearly illustrate the system's processes, user navigation, and overall operational flow in an organized and easy-to-understand manner.

Programming and Architecture Tools

- **Visual Studio Code 2026:** Visual Studio 2026 is used as the primary development environment for writing, compiling, and managing the system's code. The proponents used the platform to develop web-based applications using technologies such as React for building dynamic and interactive user interfaces, and Tailwind CSS for creating responsive

and visually consistent designs. Visual Studio Code also supports extensions and integrated tools that assist in debugging, code formatting, and real-time development, ensuring that the system functions correctly and efficiently throughout the development process.

Cloud Hosting & API Services

- **Vercel:** The proponents will utilize Vercel as the primary hosting and deployment platform for the proposed web-based reservation, time tracking, and queue management system. It allows proponents to deploy the system in a live environment where it can be accessed through web browsers on both desktop and mobile devices. Additionally, Vercel supports real-time updates and seamless deployment of system improvements, ensuring that new features and fixes can be applied without disrupting the user experience.
- **Google Gemini API:** The Google Gemini API was integrated as the primary artificial intelligence engine to power the system's wait-time prediction module. This cloud-based AI service was specifically selected over client-side machine learning libraries to accommodate the hardware limitations of the establishment, which utilizes standard consumer devices such as basic laptops and tablets. By offloading the complex predictive computations to external cloud servers via API requests, the system can dynamically analyse real-time variables such as active table sessions, party sizes, and queue lengths without consuming local CPU or memory resources, ensuring the web application remains fast and responsive during peak operational hours.

Communication & Collaboration Tools

- **Discord:** The proponents will utilize Discord as their primary communication platform. It supports collaboration through text, voice, and video channels, as well as file sharing. This platform is highly suitable for capstone projects, especially for remote teams, as it allows organized discussions via dedicated channels, real-time updates, and integration with bots or other tools that can help automate certain tasks and streamline workflows.
- **Facebook Messenger:** As an alternative communication tool, the proponents will also use Facebook Messenger to exchange messages, share files, and coordinate tasks in real time. Its user-friendly interface and widespread accessibility make it effective for quick updates and informal communication. Although it is not as versatile as specialized project

management platforms, it serves as a convenient supplementary tool for maintaining team connectivity, particularly in smaller groups.

- **Jira:** A visual project management application utilized to implement the Kanban framework for Agile Methodology development. The proponents utilized this platform to organize, track, and manage the system's development workflow efficiently. This framework ensures that the developers focus on a manageable number of active tasks at any given time, thereby improving team collaboration, maintaining transparency, and accelerating the delivery of the system's modules.

Other Software Tools

- **Microsoft Office 365:** The proponents utilized Microsoft Office 365 for the preparation and organization of all project-related documents. Applications such as Microsoft Word were used for documentation, while Microsoft PowerPoint was employed in creating and presenting project presentations throughout the development process.
- **Google Drive:** Google Drive serves as the primary platform for storing, managing, and sharing project files, including images, documents, and other resources essential to website development. It enabled efficient collaboration among team members and provided convenient access to shared materials at any time.

Calendar of Activities

MONTH	FEBRUARY				MARCH				APRIL				MAY			
ACTIVITIES																
Title Proposal																
Chapter 1: Introduction																
Project Context																
Purpose and Description																
Objectives of the Study																
Scope and Limitations																
Review of Related Literature and Studies																
Chapter 2: Methodology																
Technical Background																
Technologies to be Used																
Calendar of Activities																
Resources																
Requirement Analysis																
Requirements Documentation																
Design of Software, System, Product, and/or Processes																
References																
Designing UI/UX Period																
Consultation																
PowerPoint Presentation																
Research Documentation																
Group Meeting																
Mock Defense																

Table 4. *Calendar of Activities*



Pending



On-Going



Done

Proposed Calendar of Activities Phase 2

MONTH	JUNE				JULY				AUGUST				SEPTEMBER				OCTOBER				NOVEMBER			
ACTIVITIES																								
Software Development Testing																								
Alpha and Beta Testing for Admin Portal																								
Alpha and Beta Testing for Staff Portal																								
Alpha and Beta Testing for Customer Portal																								
Alpha and Beta Testing for AI-Powered Wait-Time Estimation																								
Chapter 3: Result and Discussion																								
Testing																								
Description of Prototype																								
Implementation Plan																								
Implementation Result																								
Conclusion																								
Recommendations																								
PowerPoint Presentation																								
Consultation																								
Appendices																								
Mock Defense																								
Final Defense																								

Table 5. *Proposed Calendar of Activities (Phase 2)*



Legend Pending



Legend On-Going



Legend Done

Requirements Analysis

The Online Reservation, Time Tracking, and Queuing System with AI-Powered Wait-Time Estimation for One Shot Bar and Billiards is designed to modernize the establishment's current manual operations, which involve recording customer information, arrival times, and booking details on paper and relying on memory to monitor order arrivals. The system provides a centralized and automated platform for efficiently managing reservations, table usage, customer queues, and order tracking.

The system will be utilized by four primary user groups: Admin Users, Staff Users, and Customer Users. Admin Users and Staff Users have access to the system's management features. Their responsibilities include handling reservations, monitoring table usage, managing queues, and tracking session time. They are also responsible for ensuring that customer data and reservation details are accurately recorded and updated within the system. Customer Users, on the other hand, will interact with the customer-facing web interface to check table availability and submit reservation requests online.

The system will streamline key business processes such as reservation management, queue handling, and table monitoring. It replaces the current paper-based and manual tracking methods with automated features including real-time table monitoring, digital queue management using a First Come, First Served (FCFS) approach, live availability dashboard, and AI-powered design assistant. These features ensure accurate tracking of customer sessions, fair queue ordering, and efficient table allocation, especially during peak hours.

The system will be implemented within the operational environment of the establishment where reservation and monitoring activities are performed. It will be deployed as a web-based platform, accessible through desktop computers and mobile devices using standard web browsers. The system requires a stable internet or local network connection to maintain its real-time features, particularly for queue updates and table monitoring.

The implementation of the system will be aligned with the operational needs of One Shot Bar and Billiards, ensuring that it is ready for use during peak hours. Prior to full deployment, testing and staff training will be conducted to ensure smooth adoption. Continuous improvements and updates

will be applied based on system performance to maintain efficiency and reliability.

Requirements Documentation

The requirements documentation defines the functional and system requirements necessary to support the operations of One Shot Bar and Billiards. These requirements were identified based on the establishment's existing manual processes, operational challenges, and the proposed features outlined in the project proposal. The purpose of this documentation is to ensure that the proposed web-based reservation, time-tracking, and queue management system effectively addresses the establishment's need for accurate scheduling, organized customer flow, and reliable monitoring of table usage. The system is intended to replace paper-based reservation records and manual monitoring procedures with a centralized platform that enables staff to manage reservations, queues, and table sessions in real time.

1. User Types and Roles

Admin User:

- Access to administrative tools for managing users, tables, promotions, system settings, bookings, announcements, calendar closures, and analytics.
- Able to view system-wide activity and export them as CSV reports.
- Able to customize the customer-facing homepage of the website.
- Able to add, edit, and deactivate staff accounts.
- Able to generate, activate, and delete promo codes.
- Able to add, activate, and deactivate tables.
- Able to edit hourly rates, happy hour schedules, overtime rates, and adjust reservation down payment percentages.
- Able to define and enforce booking rules, including minimum and maximum booking hours, party size limits, and the cancellation notice period.
- Can view system-wide analytics, table performance, and revenue usage reports.
- Can mark specific dates as closed on the calendar to prevent booking.
- Able to view feedback messages from customers.

Staff User:

- Access to reservation tools such as adding, verifying, checking in, completing, and canceling reservations.
- Access to table monitoring tools to seamlessly start, extend, or end active table sessions.
- Able to register walk-in customers into the First-Come, First-Served (FCFS) queue and monitor their AI-powered wait-times estimation.
- Able to move waiting customers to available tables and automatically cancel reservations if customers are more than 15 minutes late.
- Able to view complete booking details, including the customer's name, hours booked, party size, specific table number, and payment status.
- Able to view daily, weekly, and monthly sales and table usage.

Customer User:

- Can view real-time table availability and the AI-powered wait-time estimation for the walk-in queue before deciding to book.
- Can view the required down payment amounts and available booking time slots, with a strict minimum booking duration of one hour enforced before reservation.
- Can fill in reservation details such as name, contact number, preferred date and time, duration, and party size.
- Can interact with a visual floor plan to choose a specific billiard table for their reservation or select any available table.
- Can submit a reservation by uploading a GCash down payment reference.
- Can cancel or reschedule an active reservation and provide a written reason through their account.

2. Core System Features

Online Reservation Module:

- Staff collect the customer's full name, contact number, optional email address, preferred date and time slot, reservation duration in hours, party size, and preferred table when creating a new reservation.
- The system automatically computes the total amount due based on the hourly rate and applies to a down payment requirement upon booking.
- Each reservation is assigned one of five statuses: pending, confirmed, checked-in, completed, or cancelled, which staff update as the customer progresses through their visit.
- Pending reservations require staff verification before they are confirmed, at which point the system sends an automated email confirmation containing the booking details, total amount, down payment collected, and remaining balance.
- Staff may cancel any active reservation and are required to select a reason from a defined list of options.
- A real-time summary panel displays the number of bookings for the day, reservations pending confirmation, total revenue from completed sessions, and outstanding payments across all active reservations.

Queuing Management Module:

- Staff register for walk-in customers who are waiting for a table by entering their name, contact number, party size, and optional notes.
- The queue operates on a First Come, First Served basis, with each customer's position determined by the time they were registered.
- The module displays the full list of waiting customers in order of arrival, highlights the next customer to be served, and shows how long each customer has been waiting.
- Staff can call a customer to alert them that a table is becoming available, which

moves them to a separate called list, or removes them from the queue entirely.

- The module displays a live count of available tables alongside the waiting queue.
- Upcoming reservations for the day are also visible within the module, with a visual indicator for bookings scheduled within the next 60 minutes.

Table Monitoring and Timer Module:

- Staff can start, monitor, extend, and end sessions for individual billiard tables.
- When starting a session, staff assign a customer from either the walk-in queue or the day's confirmed reservations.
- For walk-in customers, payment is collected at the end of the session.
- For reservation customers, the down payment amount settled is recorded at the start. Staff select a session duration from preset options, and the timer begins automatically once the session is started.
- The system tracks elapsed time and calculates overtime charges for any minutes beyond the booked duration.
- When ending a session, the system presents a full payment summary showing the booked charge, overtime charges if applicable, the amount already paid, and the remaining balance.
- Staff record the payment status as fully paid, partial, or unpaid, and select the payment method used: cash or GCash.
- Sessions may be extended by selecting an additional block of time, with the extension charge calculated automatically based on the hourly rate.
- A point-of-sale sidebar allows staff to add food and beverage orders to a table running bill, which are included in the total amount due when the session ends.
- Voiding an order from the bill requires admin password confirmation.

Table Usage Report Module:

- The module displays weekly revenue through an area chart, a breakdown of session

durations by percentage, peak-hour occupancy patterns across the full operating window, and a monthly revenue trend in a bar chart.

- A table performance report lists each billiard table with its total number of sessions, revenue generated, and utilization rate.
- Each table is categorized as high demand, active, moderate, or low use based on its utilization percentage.
- The report is designed to support management decisions such as adding or reducing tables based on observed usage patterns.

3. System Design and User Interface

The proposed system design for One Shot Bar and Billiards will be visualized using tools such as Figma. This will help illustrate the user journey, core functionalities, and user interface layout. Below are the storyboards and descriptions of the key sections of the system:

Admin Portal - Web Page

The Admin Portal extends the staff dashboard with additional configuration pages accessible only to accounts with admin-level access. All changes made through the admin portal take effect immediately across the system.

1. User and Table Management - Allows administrators to manage staff accounts and billiard tables.

- Admins can create, edit, activate, or deactivate staff accounts and assign roles. Admin-level access can be toggled per account.
- Billiard tables shown in the monitoring dashboard can be added, renamed, or deactivated from this page.

2. Rates, Promo Codes, and Reservation Terms - Allows administrators to configure pricing and booking rules applied across the system.

- Admins can set the standard hourly rate, happy hour rate and schedule, down payment percentage, and overtime rate. Changes apply immediately to the reservation and billing calculations.
- Promo codes can be generated with a set of discount percentages, usage limits, and expiry date. Codes can be activated, deactivated, or deleted at any time.
- Booking rules including minimum and maximum booking hours, party size limits, and the cancellation notice period can be updated through the reservation terms editor.

3. Announcements, and Site Customization - Allows administrators to publish announcements and manage the customer-facing website.

- Announcements displayed in the customer-facing header ticker can be created, activated, or removed. Each announcement is categorized as informational, promotional, a warning, or an event, and can be set with an optional expiry date.
- Admins can update the content of the customer-facing website, including the logo, homepage slideshow images, promotional banner, hero text, about page story, and contact information.

4. Analytics, Activity Log, and Closing Calendar - Allows administrators to view and manage table analytics, and system activity.

- Charts cover daily and monthly revenue trends, session length distribution, and peak-hour occupancy patterns. A table performance report lists each billiard table with its total sessions, revenue, utilization rate, and demand category.
- The activity log records all actions performed by staff and admin accounts, including reservation changes, table assignments, payments, and administrative actions. Logs can be exported as a CSV file.
- Specific dates can be marked as closed on the reservation calendar for a written reason, which prevents customers from booking on those dates.

Staff Portal - Web Page

Staff access the dashboard through a separate login page using their assigned credentials. The system detects the account role upon login and redirects accordingly. The staff dashboard covers the day-to-day operations of the establishment, including managing reservations, monitoring table sessions, and handling the walk-in queue.

1. Reservations Page - Allows staff to view, manage, and create table bookings.

- A summary panel at the top shows the day's total bookings, pending verifications, total revenue, and outstanding balances.
- The reservations table lists all bookings with customer details, status, and payment breakdown. Staff can search by name or contact number and filter by status.
- Staff can verify a pending reservation, which sends an automated email confirmation to the customer. They can also check in, complete, or cancel a reservation, and manually create a new one on behalf of a walk-in customer.
- A detail panel opens when a reservation row is selected, allowing staff to view the full booking, mark the down payment as paid, or collect the remaining balance.

2. Table Monitoring and Queue Management - Allows staff to register and manage walk-in customers waiting for a table as well as manage all billiard table sessions in real time.

- Staff register for walk-in customers by entering their name, contact number, and party size. The queue follows the First Come, First Served method based on arrival time. The waiting list is displayed in arrival order, with the next customer highlighted, and staff can call or remove a customer as needed. The page also shows available tables and upcoming reservations, with a flag for bookings arriving within the next 60 minutes.
- Tables are displayed as cards showing each table's status, session timer, customer name, and time remaining or overtime indicator. Staff can start a session by selecting a customer from the queue or confirmed reservations, extending an active session with automatic charge calculation, or ending a session and recording the final payment.

3. History and Settings - Staff can view recent reservations and queue transactions and edit account information.

- Charts cover daily and monthly revenue trends, session length distribution, and peak-hour occupancy patterns.
- A table performance report lists each billiard table with its total sessions, revenue, utilization rate and demand category.
- The Settings page allows staff to update their personal information, username, and password.

Customer Portal - Web Page

Customers access the portal through the public-facing website. Browsing the page does not require an account, but creating one allows customers to track booking history, and submit cancellations. The customer portal serves as the primary point of contact between the establishment and its clients.

1. Landing Page - Serves as the main entry point, giving customers an overview of the establishment and direct access to reservations.

- Displays a front-page slideshow, key establishment figures such as the number of tables and hourly rate, and feature highlights for the venue.
- A fixed navigation bar provides tabbed access to all sections of the page. An announcement of ticker in the header displays operational notices set by the admin. Customers may log in or register from the header.
- Once logged in, the login and register buttons are replaced by a profile button linked to account settings.

2. Reservation Page - Allows customers to select a date, complete a booking form, and submit a down payment in a two-step process.

- An interactive calendar shows date availability. Dates marked as closed by management are blocked, and past dates are disabled.
- After selecting a date, customers can view available time slots, the real-time status of all billiard tables, and a snapshot of the current walk-in queue.
- This module includes logic for handling edge cases based on the establishment's specific policies. Staff may perform an "Early Check-in" when a table is available, or place the customer next in line for the next available table if they exceed the strict 15-minute grace period.
- The booking form collects the customer's name, contact number, email address, party size, duration, preferred date and time slot, and preferred table.

- An optional promo code field allows customers to apply for a discount before submitting.
- Can view the total cost computation and choose flexible down payment options, either half or full payment, before finalizing the booking.
- After form submission, a payment modal displays the GCash QR code and account details. The customer enters their 13-digit GCash reference number and uploads a screenshot as proof of down payment. The staff validates the reference number before completing the reservation.
- Logged-in customers can view their booking history and cancel an active reservation by selecting a reason from a defined list.

3. About Us and Feedback Pages - Provides establishment information and a channel for customers to contact management.

- The About Us page displays the establishment's location on an embedded map, a list of available facilities, contact details, and social media links.
- The Feedback page contains a form where customers can submit a message to management, categorized by type such as suggestion, complaint, or lost item.

Functional Requirements

User Management

- **REQ001.** The system shall allow users to register, log in, and log out securely.
- **REQ002.** The system shall support role-based access (Admin, Staff, and Customer).
- **REQ003.** The system shall allow users to update their profile and account credentials.
- **REQ004.** The system shall allow admins to create, edit, activate, and deactivate accounts.

Customer Functionalities

- **REQ005.** The system shall allow customers to view real-time table availability with visual indicators.
- **REQ006.** The system shall allow customers to select a reservation date, time, duration, party size, and preferred table, while strictly enforcing the establishment's minimum booking duration of one hour.
- **REQ007.** The system shall allow customers to view the current walk-in queue size and the AI-powered wait-time estimation before deciding whether to book an appointment or proceed as a walk-in
- **REQ008.** The system shall allow customers to input promo codes and apply discounts.
- **REQ009.** The system shall allow customers to cancel reservations and provide a reason for the cancellation.

Queue Management

- **REQ010.** The system shall allow staff to register walk-in customers into a digital queue that strictly follows a First-Come, First-Served (FCFS) basis.
- **REQ011.** The system shall utilize an AI-powered wait-time estimation model that analyzes active table sessions to provide accurate waiting time estimates for customers in the queue.
- **REQ012.** The system shall auto cancel a reservation if they are less than 15 minutes late and should allow staff to reassign a reserved customer to the next available table if they arrive more than 15 minutes late and their originally assigned table has already been released to the walk-in queue.

Tables Monitoring and Timing

- **REQ013.** The system shall automatically start a timer when a table session begins.
- **REQ014.** The system shall notify staff when a session is nearing expiration or has exceeded paid time.

- **REQ015.** The system shall store session duration for reporting and billing references.

Reservation Management

- **REQ016.** The system shall allow staff to create, update, verify, and cancel reservations.
- **REQ017.** The system shall prevent scheduling conflicts and double bookings by enforcing strict validation of time slots and table identifiers.
- **REQ018.** The system shall allow staff to perform an “Early Check-in” when a reserved customer arrives before the scheduled time, and the assigned table is available.
- **REQ019.** The system shall automatically compute the total rental cost and allow customers to choose flexible down payment options, specifically either a 50% down payment or full payment in advance.

Payment Processing

- **REQ020.** The system shall automatically compute the total rental cost and the required down payment.
- **REQ021.** The system shall allow staff and administrators to verify GCash reference numbers and uploaded payment proofs before confirming a booking.
- **REQ022.** The system shall support bookings with fractional hour durations and shall automatically compute and apply overtime charges when a customer exceeds the paid booking period. If the overtime duration is between 10 and 15 minutes, the system shall round up the charge to the next half-hour increment in accordance with the establishment’s penalty policy. Authorized staff members shall have the capability to manually waive overtime charges in cases where the delay is caused by staff related issues or system errors.

Non-Functional Requirements

Operational Requirements

- **REQ023.** The Customer Portal shall be implemented as a responsive web application accessible through standard web browsers on both desktop and mobile devices.

- **REQ024.** The Admin and Staff Portals shall be packaged and deployed as an offline-capable desktop application installed locally on the establishment's hardware.
- **REQ025.** The staff system shall utilize a local database to ensure continuous timer tracking, queuing, and billing operations during unexpected internet service interruptions.
- **REQ026.** The system shall automatically synchronize all locally stored offline transaction data to the cloud database once an internet connection is restored.

Performance Requirements

- **REQ027.** The system shall provide real-time updates of table status, queue tracking, and availability dashboards.
- **REQ028.** The system shall support simultaneous access by multiple staff members without data conflicts.

Security Requirements

- **REQ029.** The system shall restrict all administrative functions, configuration settings, and billing void operations to authorized administrator accounts only.
- **REQ030.** The system shall store customer reservation data securely to prevent unauthorized access.

Cultural and Political Requirements

- **REQ031.** The system will work based on how businesses operate in the Philippines. This means it will follow local business hours, support GCash payments, and consumer interaction norms.
- **REQ032.** The system shall be implemented in English, which is the primary language used in the establishment's operations and documentation.
- **REQ033.** The system shall follow to Philippine data protection and data handling

regulations applicable to local establishments that store customer information.

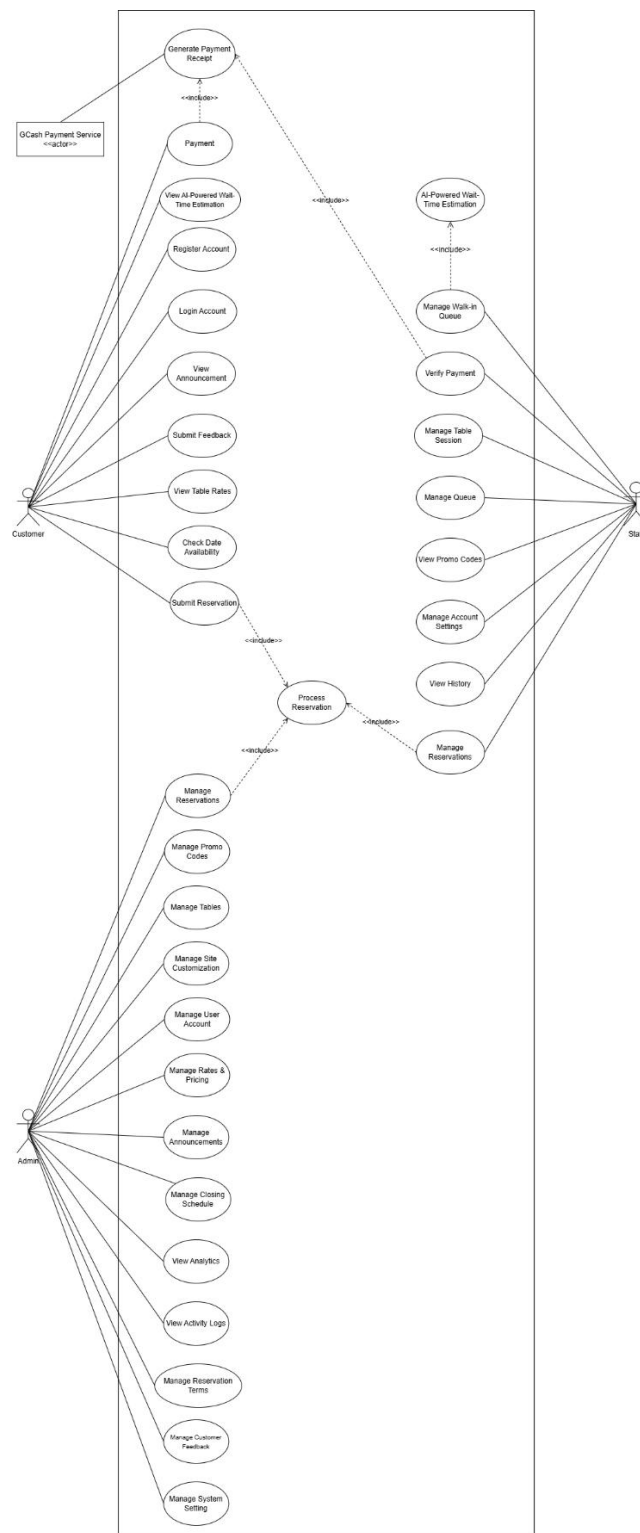


Figure 3. Proposed use-case diagram for Admin, Staff, and Customer actions

Design of Software, System, Product, and/or Processes

System Architecture

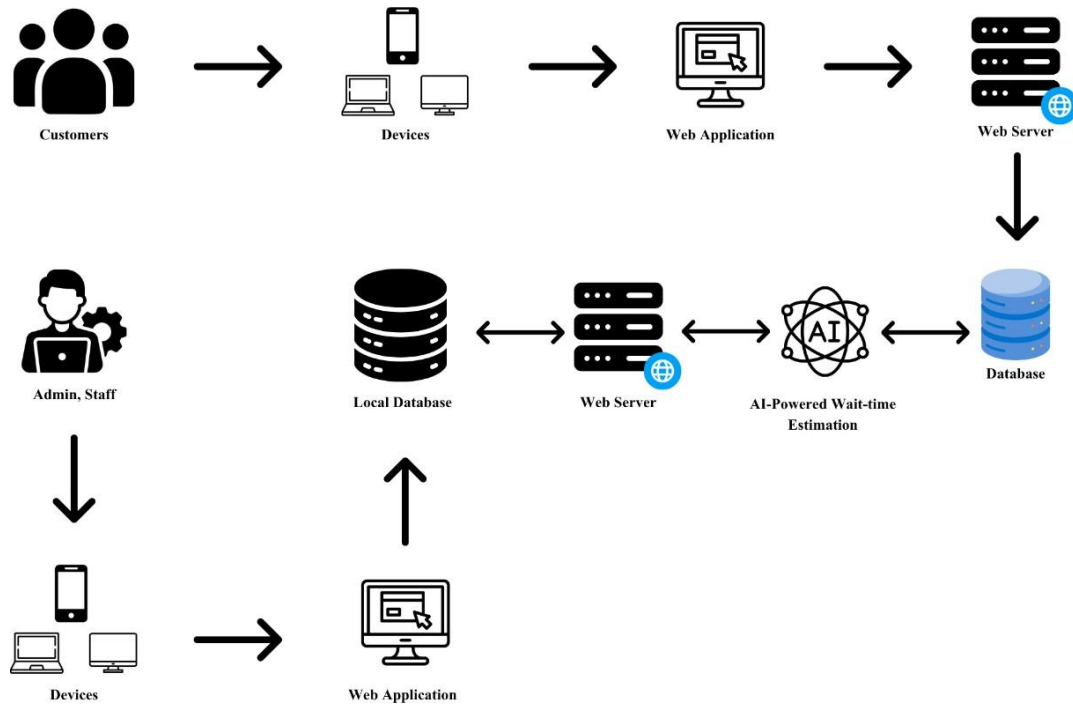


Figure 4. *Proposed Project Framework Diagram*

This section presents the dual-platform system architecture and its technical workflow, illustrating how data flows between users, the local environment, and the cloud infrastructure.

The upper sequence represents the Customer Web Portal. Customers access the system through personal devices such as mobile phones or laptops using a standard web browser. Their interaction with the cloud database is restricted and highly controlled. They can only retrieve read-only information, such as live table availability and queue status, and submit limited write requests, including reservation forms and payment references, directly to the cloud server. This restricted access ensures that customers cannot directly modify, intercept, or delete critical establishment data.

The lower sequence illustrates the Admin and Staff Portal, which is designed with an offline-capable architecture to ensure operational resilience. Instead of accessing the system through a

standard web browser, staff members use an Electron-based desktop application installed directly on the establishment's hardware. The application continuously interacts with a local database, enabling staff to perform full Create, Read, Update, and Delete (CRUD) operations on active table timers, queue sequences, and billing computations even without an internet connection. When a stable internet connection becomes available, the desktop application utilizes a two-way synchronization protocol to seamlessly synchronize local data with the cloud database. This process ensures that all records remain accurate and consistent across the entire system while minimizing the risk of data loss or service interruptions.

Conceptual Design

This section explains how the project requirements were identified based on the daily operations of One Shot Bar and Billiards. It includes the system specifications and the design of the proposed online reservation, time tracking, and queuing system with AI-powered wait-time estimation. To better understand the needs of the establishment, the proponents worked closely with the administrator and staff of One Shot Bar and Billiards to identify the problems currently encountered by the business and determine its specific operational needs. This collaborative approach between the proponents and the establishment ensures that every aspect of the developed system directly addresses the existing problems of the business and meets the expectations of the management by providing a more modernized and efficient solution.

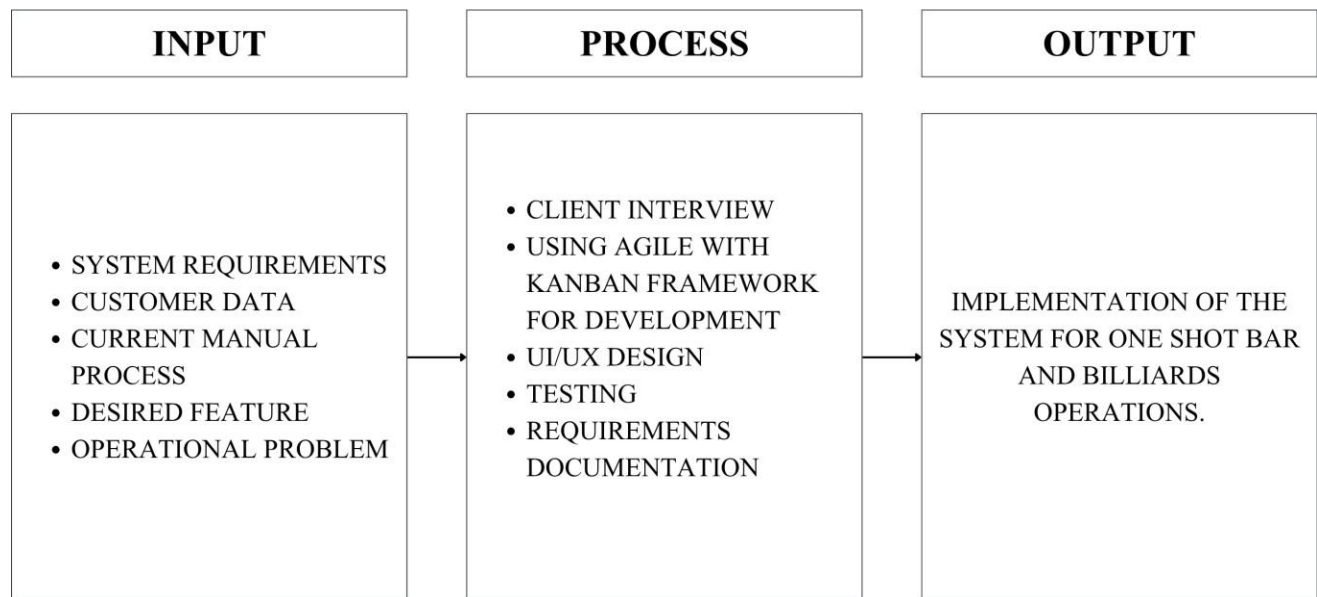


Figure 5. *Proposed Conceptual Framework Diagram*

The diagram shows the Input-Process-Output (IPO) framework used in the development of the system for One Shot Bar and Billiards. The input section contains the necessary information needed for the development of the system, such as the system requirements for One Shot Bar and Billiards, customer data to be stored, their current manual processes, the features they want to include, and the operational problems that led to the development of the project.

The Processing Section refers to the steps and procedures involved in the development of the system. This includes interviewing the management to understand their operations and

requirements, applying the Agile with Kanban framework methodology for incremental system development, designing the UI/UX, conducting alpha and beta testing with administrators, staff, and customers, and documenting all functional and system requirements.

The Output section presents the final system for One Shot Bar and Billiards, which covers all the required features and functions to be implemented in the establishment's operations. The designed system aims to improve the establishment's workflow and operational performance while enhancing the reservation and queuing process for customers, resulting in a better overall experience for both staff and customers.

Requirement Analysis

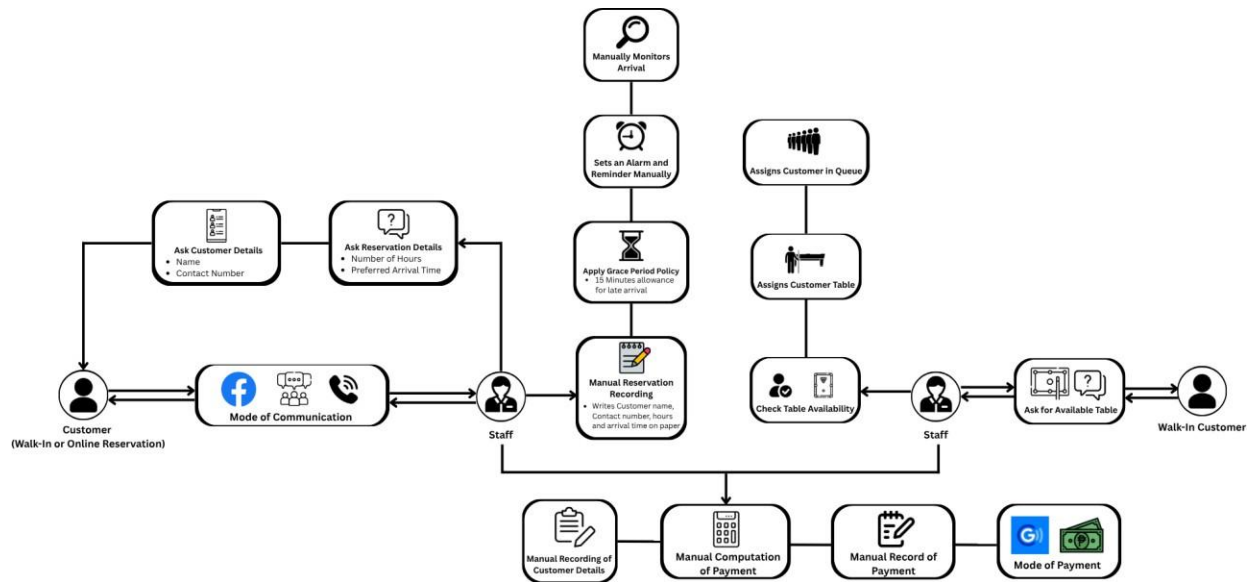


Figure 6. *Current Technical Procedure of One-Shot Bar and Billiards*

One Shot Bar and Billiards currently manage reservations, walk-in customers, and payment transactions through direct staff interaction using manual processes. Customers who wish to reserve tables either walk in or contact the staff through Facebook Messenger, group chats, or phone calls to provide reservation details such as their preferred arrival time, number of hours, name, and their desired table. Staff manually check table availability before confirming reservations and recording customer information on paper. To monitor reservations, staff manually set alarms and reminders while also implementing a 15-minute grace period policy for customers who fail to arrive on time. If customers exceed the allowed grace period, the reservation is cancelled to prevent the loss of table availability. Walk-in customers also rely on staff to manually check available tables before being assigned to a table or placed in a queue if tables are unavailable. Customer details, payment computation, and payment records are all processed manually, making the overall operation difficult to organize and monitor efficiently, especially during peak hours when multiple customers and reservations are being handled simultaneously.

Adopting an integrated reservation and management system would improve operational efficiency for both customers and staff by automating reservation handling, queue management, and transaction processing. Customers would be able to check table availability and make reservations

more conveniently without relying entirely on manual staff process, while staff would benefit from automated monitoring of reservations, grace periods, table usage, and customer records. Automating tasks such as time tracking, payment computation, and record-keeping would reduce manual workload and minimize delays during peak operating hours. Overall, the proposed system would provide a more organized and reliable process for handling reservations and walk-in customers while improving the overall customer experience. During the initial stages of development, the required processes and workflows were identified and organized, and data flow diagrams were created to clearly represent the interactions, processes, and manual data handling currently practiced within the One-Shot Bar and Billiards.

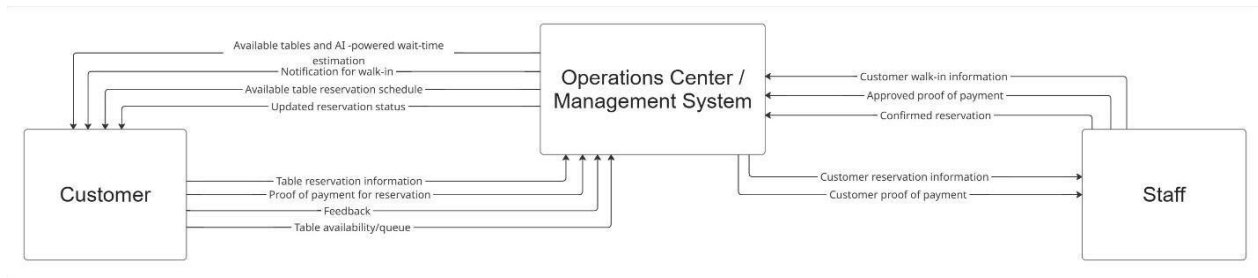
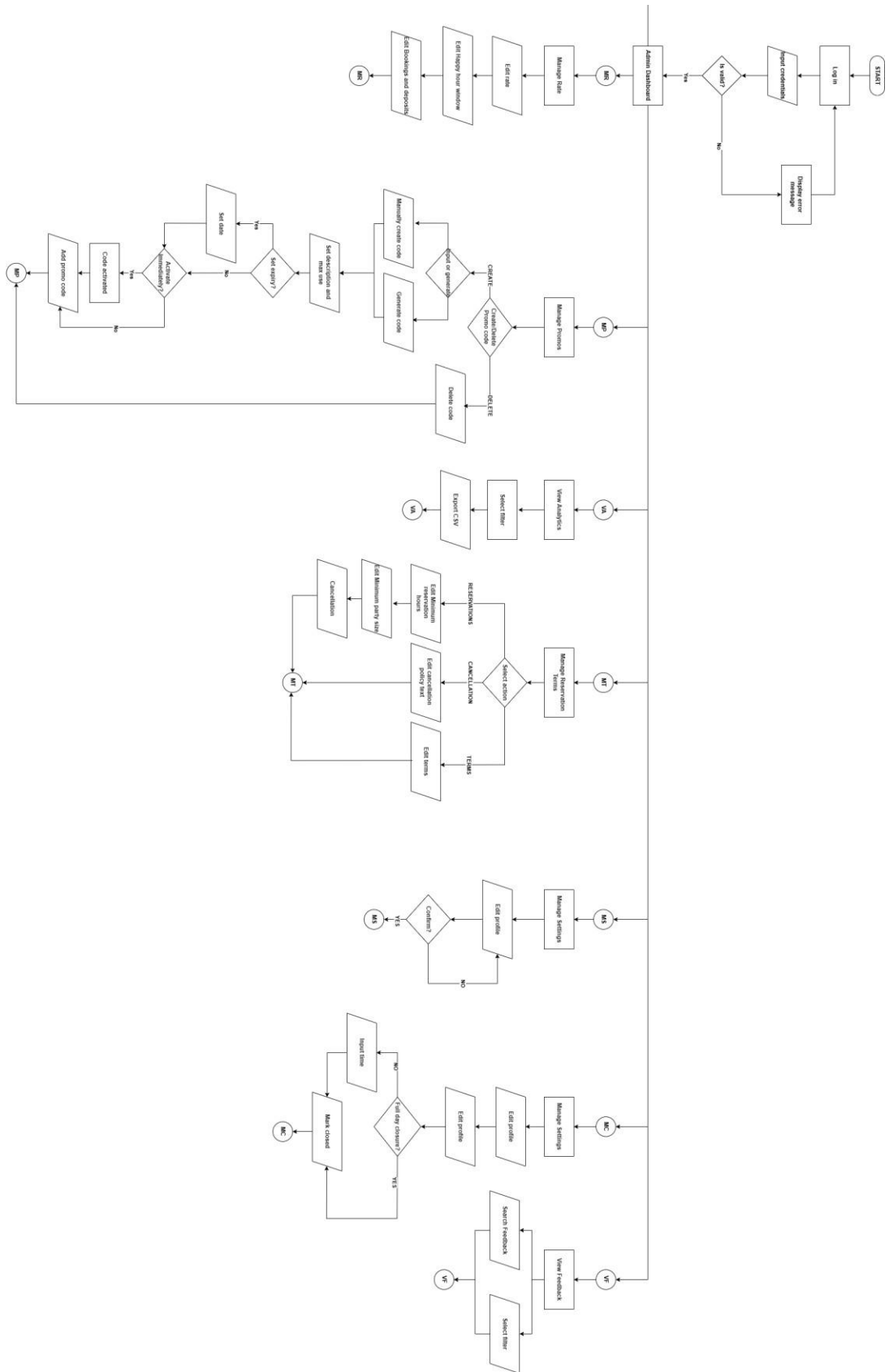


Figure 8. *Proposed Data Flow Diagram for Customer*

The diagram illustrates how the Admin and Staff roles interact with the Operations Center / Reservation Management System. The admin is responsible for all system configuration, sending inputs such as site customization settings, user account data, table records, promo codes, hourly rates, happy hour schedules, reservation terms, and closing calendar entries into the system. In exchange, the system returns analytics reports, activity logs, compiled customer feedback, and daily reports, giving the admin a way to monitor operational performance without being involved in each transaction. Staff handle the day-to-day operational side. They record customer table sessions, reservation sessions, customer queue data, walk-in customer details, and can update their own account settings through the system. In return, the system provides them with the updated queue sequence and AI wait-time, current reservation list, activity history, available promo codes, and daily reports for reference during service. This keeps staff working from a live, centralized record rather than coordinating through verbal instructions or separate handwritten notes.



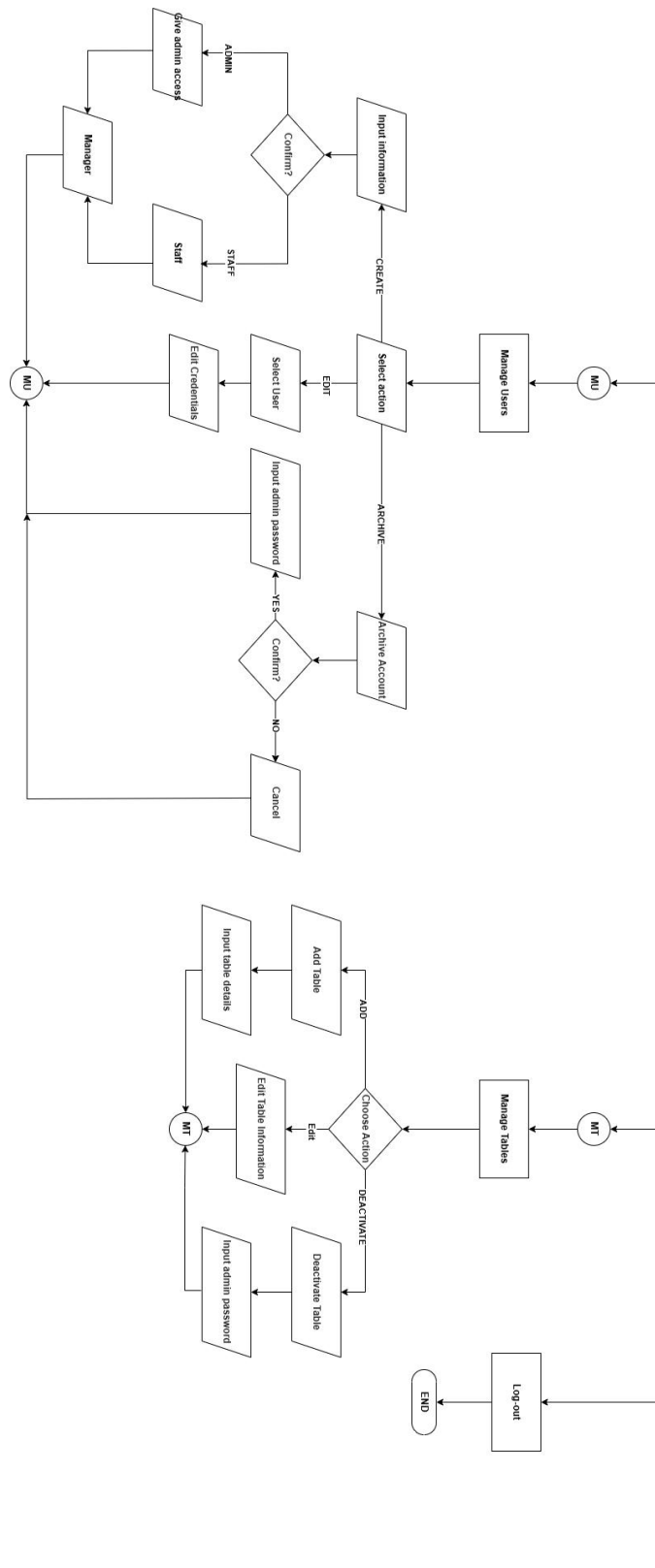


Figure 9. Flowchart for Admin

The admin flowchart illustrates the overall administrative process involved in managing users, tables, reservations, promo codes, analytics, feedback, and system configurations within the reservation and management system. The process begins with the login procedure, where the administrator enters account credentials to access the system. The system validates the entered credentials and determines whether the login information is correct. If the credentials are invalid, the system displays an error message and returns the administrator to the login page. Once the credentials are verified successfully, the administrator gains access to the Admin Dashboard, where different management modules and administrative functions can be accessed.

One of the main functions available in the Admin Dashboard is user management. In this process, the administrator can manage staff and administrator accounts by selecting different actions such as creating, editing, or archiving accounts. When creating a new account, the administrator inputs user information and confirms whether the account should be assigned staff or admin access privileges. The system then grants the selected access level and stores the account information. For existing accounts, the administrator may edit user credentials or archive accounts by entering the admin password for confirmation. The system validates the entered password before proceeding with account archiving. If the confirmation fails, the process is canceled and returned to the management section. This module ensures proper account control and user access management within the system.

The flowchart also includes table management and rate management processes. In table management, the administrator may add new tables, edit existing table information, or deactivate unavailable tables. Adding a table requires the administrator to input table details, while editing allows modifications to existing table information. If the administrator chooses to deactivate a table, the system requires password confirmation before completing the process. These features allow administrators to maintain accurate and updated table availability information. In rate management, administrators can manage service rates by editing rates, adjusting happy hour sessions, and modifying booking or deposit settings. This ensures that pricing structures and reservation policies remain updated according to business requirements.

Another important section of the admin flowchart is the promo management and reservation management process. In promo management, administrators can create, edit, activate, or delete

promo codes used for customer reservations. Promo codes may either be generated automatically or created manually. After creating the promo code, the administrator configures details such as discount value, maximum usage limits, and activation settings. The system checks whether the promo code setup is complete before activating the code. If activation is successful, the promo code becomes available for customer use. The reservation management section allows administrators to manage reservations, cancellations, and terms or conditions related to bookings. Administrators may edit minimum reservation hours, adjust cancellation policies, and modify reservation terms to ensure that booking operations follow business rules and scheduling requirements.

The flowchart also presents analytics, feedback, and settings management functions. Through the analytics module, administrators may apply filters and export reports in CSV format for monitoring system performance, reservations, and operational activities. The feedback section enables administrators to search customer feedback and apply filters for easier review and monitoring of customer concerns or suggestions. In the settings module, administrators may edit profile information and configure operating schedules. The system allows administrators to define whether the establishment is under full-day closure or partial closure by entering specific operating times. The system then updates the closure settings accordingly. Additionally, administrators can manage profile information by editing and confirming profile details before saving changes to the system.

The admin flowchart concludes with system maintenance and account control functions that support the overall operation of the establishment. Administrators are responsible for managing operational data, maintaining user permissions, controlling reservation settings, and ensuring that promo codes, rates, and schedules are properly configured. The process ends when the administrator logs out of the system after completing the required administrative tasks. Overall, the admin flowchart illustrates a centralized and organized administrative workflow that supports system configuration, operational monitoring, user management, reservation policy control, and business management functions within the reservation and service management system.

The staff flowchart illustrates the process followed by staff members when managing customer reservations, table monitoring, transactions, records, and system operations within the reservation and management system. The process begins with the login procedure, where staff members enter their username and password through the login page. The system validates the entered credentials and displays either an error message for invalid credentials or grants access to the Staff Dashboard upon successful authentication. Once logged in, staff members can navigate through several operational modules, including table monitoring, queue management, reservation management, customer reservation handling, transaction monitoring, history records, promo code management, and system settings.

The table monitoring process allows staff to monitor and manage billiards table usage in real time. Staff members may view table status and select a session to start, extend, or end. When starting a session, the staff selects customer details, chooses session duration, calculates payment amounts, and activates the customer session. Once the session begins, the system displays a confirmation message and continuously updates session duration information. Staff may also extend an active session by selecting additional duration, recalculating charges, processing payment, and updating session records. If the session duration exceeds the allowed time, the system displays a session expiration message. Staff members can also end sessions, release table allocation, and return the table to available status after confirming customer checkout and validating payment details.

The flowchart also includes the queue management and reservation management processes. In queue management, staff members monitor the current queue list, view available tables, and determine whether tables are vacant or occupied. The system also integrates an AI-powered wait-time estimation feature designed to provide highly accurate waiting time predictions for queued walk-in customers. Instead of relying on manual staff estimations, the proposed system uses AI-powered wait-time estimation to analyze previous operational data such as time of day, customer party size, average session duration, active table sessions, occupied tables, and upcoming online reservations. By cross-referencing these data, the system provides more accurate and clear wait-time estimates that help reduce customer confusion to minimize customer walk-outs, and improve queue management efficiency during peak operational hours. If a table becomes available, the system allows staff to assign customers from the waiting queue to the available table. Staff may also call customers from the queue, sort the waiting sequence, update queue information based on

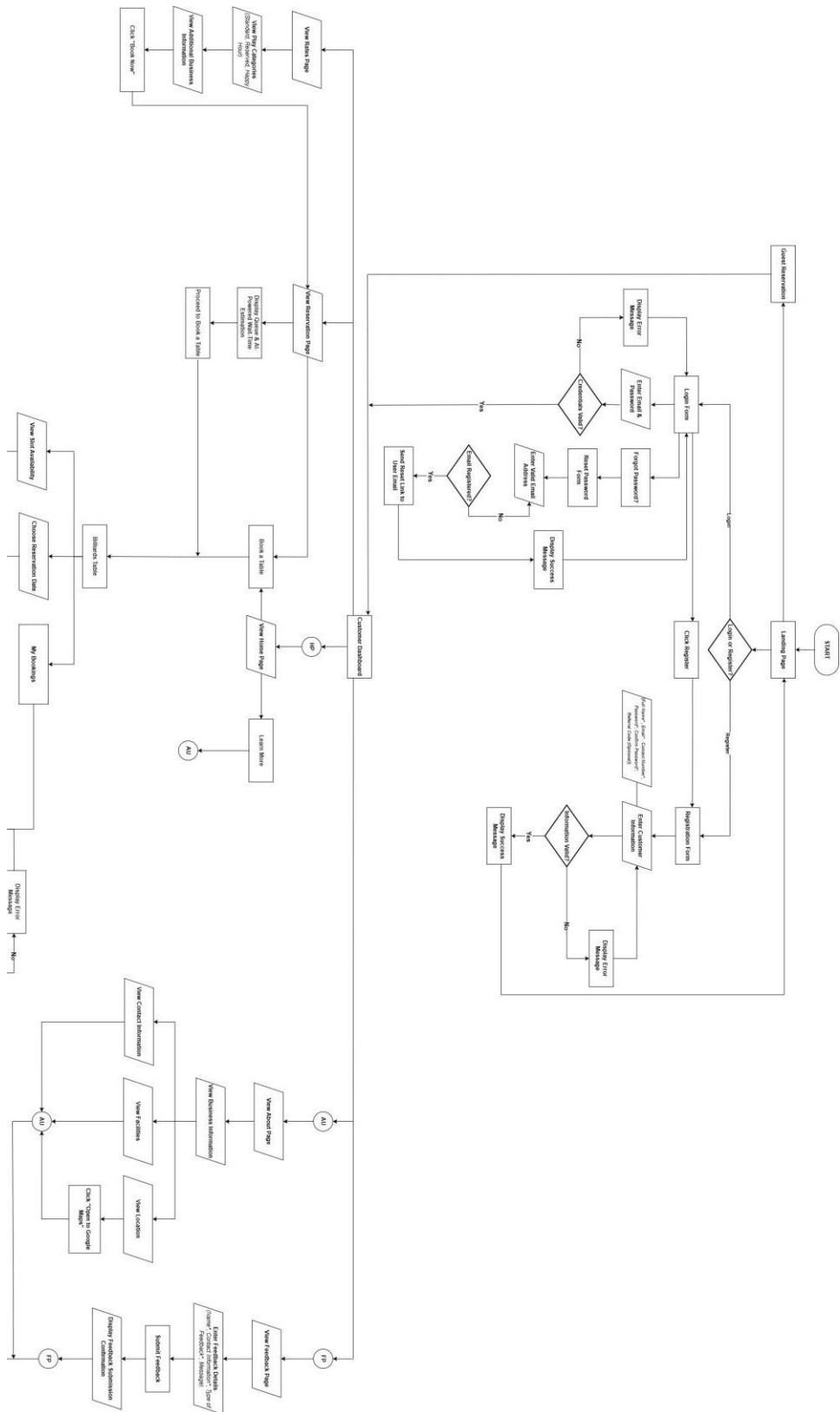
customer status, and display AI-powered wait-time estimation for newly queued customers. In reservation management, staff members can view dashboard summaries, check reservation lists, monitor reservation schedules, and access reservation details. Staff may also review table status and table monitoring records to ensure that reservations and active sessions are properly coordinated within the system.

The customer reservation handling process allows staff to manage reservation requests submitted by customers. Staff members may display reservation overviews, view customer records, search or filter reservation information, and access customer receipts. Once the reservation is verified, staff can either accept, reject, or cancel the booking request. If approved, the system processes the customer reservation and activates the customer session. If rejected or canceled, the system requires the staff to provide a reason for cancellation before confirming the cancellation request. This process ensures that customer reservation requests are properly monitored and validated before approval.

The staff flowchart also covers transaction monitoring, history record management, and promo code management. In transaction monitoring, staff members can view transaction history, monitor table occupancy, track walk-in customers, and view ongoing table usage lists. Staff may also access booking and transaction details for customer monitoring purposes. The history records section allows staff to display history records, search or filter transaction and booking data, and manage record actions. Staff members can view records, delete selected history records, or send data history requests for approval. The system validates whether an action is allowed before displaying either an error message or a successful action confirmation. Additionally, the promo code management module enables staff to view promo codes, display promo details, generate promo links, and configure discount settings such as average discount values and total usage limits for promotional offers.

The flowchart includes the settings and account management features available to staff members. Through the settings section, staff may configure system settings, display profile information, update profile details, and modify security credentials. Staff members may edit profile information by entering updated personal details, while account credentials such as usernames and passwords can also be updated through credential verification. The system validates the submitted

information before displaying either a success message or an error message. After completing the requested updates, staff members are returned to the settings menu. The process concludes when the staff member logs out of the system. Overall, the staff flowchart presents a complete operational workflow that supports reservation monitoring, AI-powered wait-time estimation, table management, queue handling, transaction processing, customer verification, record management, promo configuration, and system maintenance within the establishment.



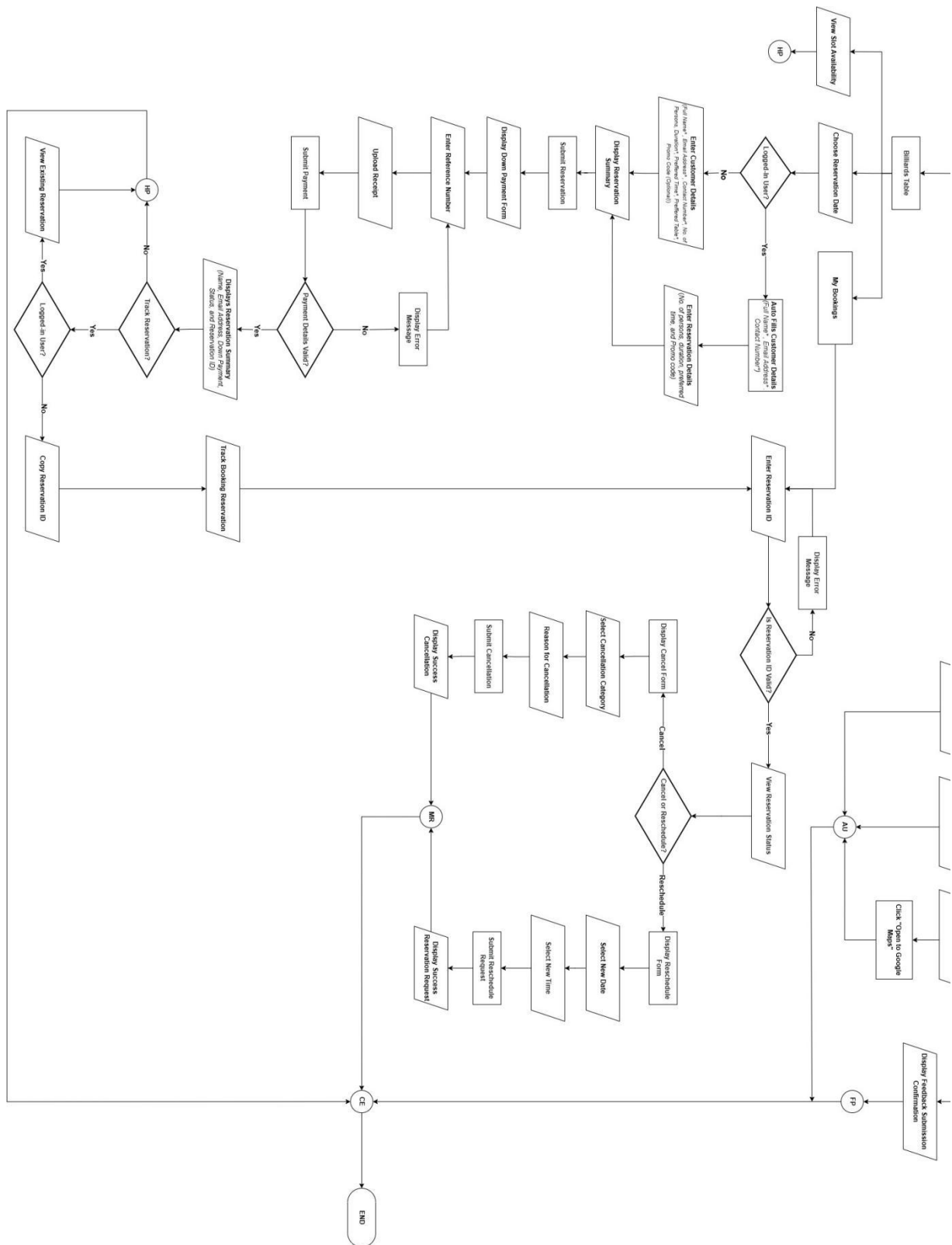


Figure 11. Flowchart for Customer

The customer flowchart illustrates the complete process followed by customers when interacting with the reservation and management system. The process begins at the landing page, where customers are given the option to log in, register a new account, or continue as a guest reservation user. If the customer chooses to register, they are directed to the registration form where they must provide customer information such as full name, email address, password, and contact details. The system validates the entered information and displays either a success message or an error message depending on whether the submitted details are valid. Existing users may log in by entering their email and password through the login form. If incorrect credentials are entered, the system displays an error message. Customers who forget their password may access the reset password form, where they enter their registered email address to receive a reset link through email confirmation. After successful login or registration, the customer is directed to the Customer Dashboard.

From the Customer Dashboard, customers can navigate to several sections of the system, including the Home Page, Reservation Page, About Page, Feedback Page, and Rates Page. The Home Page allows customers to learn more about the business and proceed with booking a table reservation. The Rates Page displays the available play categories such as standard, reserve, and happy hour rates, together with additional business information that helps customers understand the available services before proceeding with a reservation.

The reservation process starts when the customer selects the billiards table reservation feature. Customers may first view slot availability and choose their preferred reservation date. The system also integrates an AI-powered wait-time estimation feature to provide highly accurate waiting time predictions for queued walk-in customers. The system uses AI-powered wait-powered wait-time estimation to reduce inaccurate wait-time estimates by analyzing previous operational data such as time of day, customer party size, average session duration, active table sessions, occupied tables, and upcoming online reservations. Instead of relying on rough staff estimations, the AI-powered wait-time estimation feature provides more accurate and clear wait-time estimates that allow customers to better manage their expected waiting time before proceeding with reservations or queuing. This helps reduce customer uncertainty, minimize customer frustration, and decrease the possibility of customer walk-outs during peak business hours. The system then checks whether the customer is a logged-in user. If the customer is logged in, their saved details such as full name, email address, and contact number are automatically filled in by the system. If the customer is not

logged in, they are required to manually enter their customer details together with reservation information such as preferred date, preferred time, and promo code. After entering the required information, the system displays the reservation summary for customer review before submission. Customers then proceed to the payment section, where they are shown the down payment form. They are required to enter the payment reference number, upload the payment receipt, and submit the payment details for validation. If the payment information is invalid, the system displays an error message. If validated successfully, the system generates and displays the reservation summary, including the customer's name, email address, reservation status, and reservation ID.

The flowchart also includes the reservation tracking and booking management process. Customers may track their reservation using the reservation ID provided after successful booking. The system validates whether the reservation ID is correct before displaying the reservation status. Logged-in users may also access existing reservations directly through the “My Bookings” section. Once the reservation status is displayed, customers are given the option to either cancel or reschedule their reservation. If the customer chooses cancellation, they are required to select a cancellation category and provide a reason for cancellation before submitting the request. The system then displays a successful cancellation confirmation. If the customer chooses rescheduling, the system displays the reschedule form where the customer selects a new date and new time before submitting the request. After successful submission, the system displays a reservation request confirmation message.

The flowchart also includes customer support and information-related features such as the About Page and Feedback Page. The About Page allows customers to view business information, contact information, facilities, and business location details. Customers may also access Google Maps directly through the location feature for easier navigation to the establishment. Meanwhile, the Feedback Page enables customers to submit feedback by entering details such as their name, contact information, feedback type, and message. After submitting feedback, the system displays a confirmation message indicating successful feedback submission. Overall, the customer flowchart presents a complete and organized customer journey, covering account access, reservation management, AI-powered wait-time estimation, payment processing, booking tracking, cancellation and rescheduling requests, and customer feedback submission within the system.

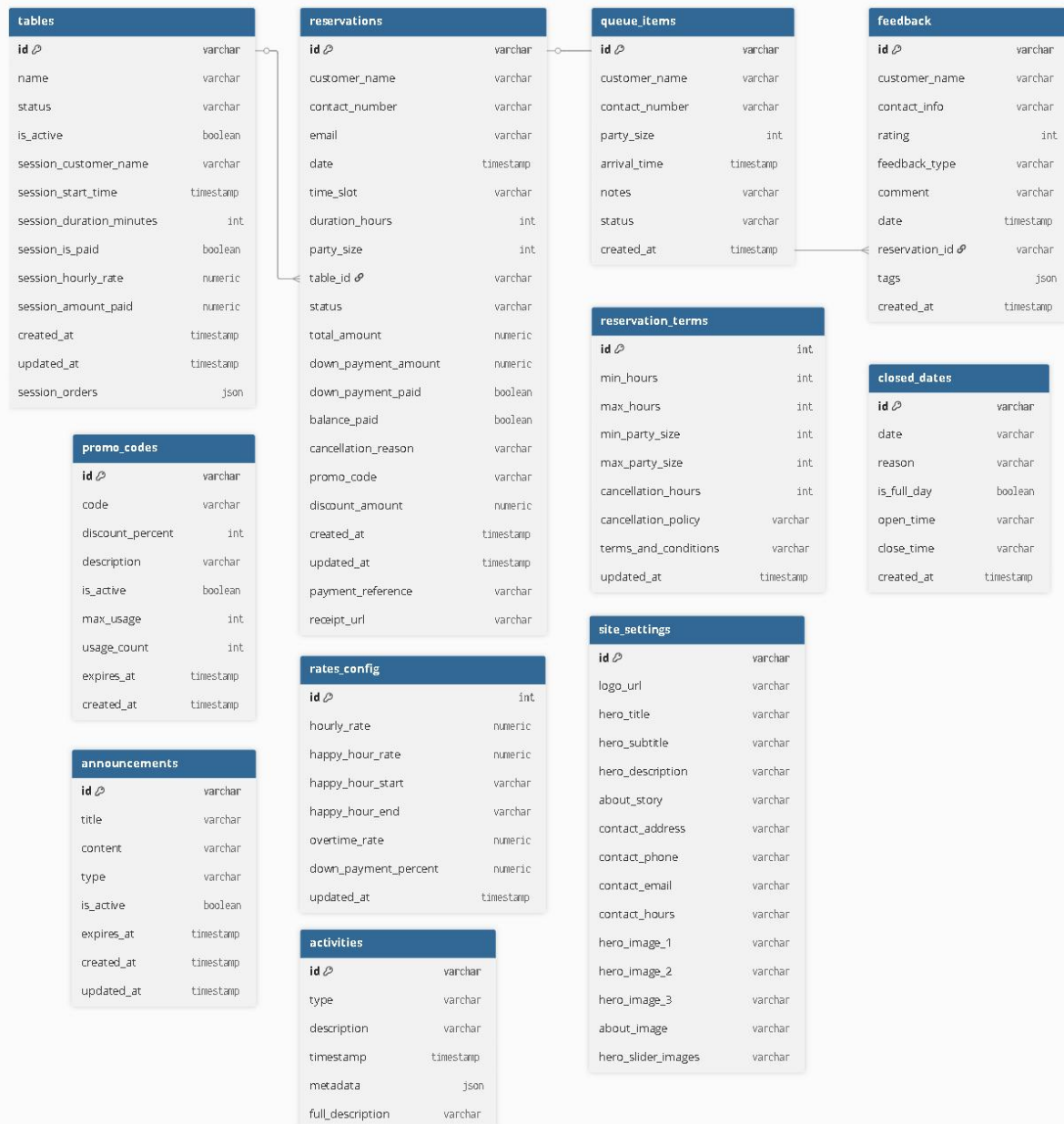


Figure 12. Entity Relationship Diagram

This section presents the Entity Relationship Diagram (ERD) of the database structure for the online reservation, time-tracking, and queuing system designed for One Shot Bar and Billiards. It illustrates the various database tables and their interconnections, each serving a specific purpose in supporting the system's core functionalities.

The core operational data is managed through three interconnected tables. The **tables** entity manages the nine billiard tables within the establishment, storing each table's current session status, active customer, hourly rate, and payment information. This is directly linked to the **reservations** table, which handles advance and walk-in bookings for billiard sessions. To support the system's interactive floor plan feature, the reservations table utilizes a foreign key (**table_id**) to assign customers to specific tables. Additionally, this table records customer details, party size, session duration, exact payment breakdowns, and promo code usage. The reservations table is further connected to the **feedback** table, tying specific customer reviews and ratings back to their original booking records.

Several supporting configuration tables manage the system's automated rules and secondary operations. The **promo_codes** table stores active discount codes, tracking their usage limits and expiration dates. The **rates_config** table houses the dynamic pricing variables, including standard hourly rates, happy hour schedules, and overtime charges. Business policies—such as the strict one-hour minimum booking, maximum party size limits, and cancellation rules—are enforced through the **reservation_terms** table.

Walk-in customers are managed by the **queue_items** table, which tracks arrival times, group sizes, and waiting status to supply data to the AI wait-time predictor. Furthermore, the **closed_dates** table records operational closures, while the **announcements** table manages public notices displayed on the customer portal. The **activities** table serves as an audit trail, securely logging important staff actions and system events. Lastly, the **site_settings** table stores front-end content such as banner images, contact details, and layout settings. Overall, the ERD visually demonstrates how these structured datasets interact to ensure a seamless, data-driven workflow for the establishment.

Breakdown of Expenses

Quantity	Specifics	Actual Cost
N/A	N/A	N/A

Table 6. *Breakdown of the Expenses*

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APPENDICES

APPENDIX A. TRANSCRIPT OF INTERVIEW

Interview Transcription of Mr. Eissen Justine

Interviewer: Arnoco, Zeji S.

De Guzman, Karl S.

Grace, Michael H.

Morcozo, Mark Joseph D.

Interviewee: Eissen Justine Aya-ay

Morcozo: Ilan po ang current tables nyo ngayon?

Mr. Aya-ay: 7 na regular, 2 vip. Dun sa tapat yung vip section namin.

Morcozo: Ilan po usually yung customers per day?

Mr. Aya-ay: Siguro tantsahin nalang natin, kasi nagru-run ito ng 12 hours. Sabihin mo ng lahat ng tables available per oras, 9x12 mo iyon. Bali 108 tables ang kaya naming ma-accommodate per day.

Morcozo: May online reservation po kayo ngayon?

Mr. Aya-ay: Meron pwede naman kayo mag-chat sa amin or sa page.

Morcozo: Pano po yung process ng online reservation?

Mr. Aya-ay: Magc-chat lang kayo sa page or kahit sakín nalang.

Morcozo: Pano po niyo nililista yung mga reservation?

Mr. Aya-ay: Naka alarm nalang sa amin yun, may notes kami kung sino yung mga naka reserve.

Morcozo: Paano po kapag dumating yung customer ng mas maaga sa reservation time nila, paano niyo po sila ina-accommodate?

Mr. Aya-ay: Pwede naman sila pumunta ng mas maaga sa reservation time nila as long as merong available na table. Kapag walang available na table, kailangan nilang maghintay sa oras ng reservation nila.

Morcozo: Paano naman po kapag hindi sumipot at di na nakapag reply?

Mr. Aya-ay: Kapag no show, matik binibigay nalang agad naming sa iba yung table na pina-reserve nila. Para hindi naman din kami malugi.

Morcozo: Anong oras po yung operation hours niyo?

Mr. Aya-ay: 12PM to 12AM

Morcozo: Anong oras po yung pinaka-maagang nagre-reserve yung customer and anong oras din po yung pinaka late?

Mr. Aya-ay: Pinaka-maagang nag-rereserve 4PM tapos onwards na iyon, tuloy tuloy na.

Morcozo: Kapag natapos na yung isang customer na naglalaro, kailangan ba ng preparation time or cleaning time bago mag-umpisa yung next reservation?

Mr. Aya-ay: Lilinis lang namin yung table tapos magwa-walis, kasi kailangan maayos siya tingnan para sa mga customers para magustuhan nila.

Morcozo: Mga ilang minutes po inaabot yung cleaning?

Mr. Aya-ay: Kahit mga 2 minutes kaya na.

Morcozo: Possible po ba magpa-reserve ng dalawang table or more yung isang tao at the same time?

Mr. Aya-ay: Yes, pwede. Ibibigay ko sa kanya iyon pero running time na iyon. Kumbaga kung anong oras yung reservation time niya, tuloy tuloy na iyon.

Morcozo: Paano po kapag dalawang tables yung pina-reserve nila, pero nung dumating sila, isa nalang yung kinuha or ginamit nila, babayaran pa rin po ba nila yung dalawa or isa nalang?

Mr. Aya-ay: Hindi na namin pinapabayaran, magiging available nalang yung isang table na hindi nila kinuha.

Morcozo: Ano po yung minimum and maximum time ng customers for reservation?

Mr. Aya-ay: Pwede naman isang oras pero normally mga nag-oopen time sila.

Morcozo: Pwede ba sila magreserve ng 30 minutes lang?

Mr. Aya-ay: Wala kaming 30 minutes kapag starting hours, kumbaga kapag magsi-simula ka ng laro, 1 hour na talaga. Ang 30 minutes namin is for extension lang. Kunyari natapos na nila yung isang oras, dun na sila pwede mag 30 minutes lang.

Morcozo: Anong oras po yung peak hours niyo ngayon?

Mr. Aya-ay: Usually mga gabi, mag-uumpisa sya 7PM to 8PM.

Morcozo: Nalo-loose track niyo po ba minsan yung pila kapag during peak hours?

Mr. Aya-ay: Oo, Hindi mo na namamalayan kung sino yung mga unang dumating tsaka sumunod.

Morcozo: Nakaranas na po ba kayo ng mga errors kapag may nagwa-walk-in sa pila like nauna po siya, pero nahuli siya sa pila?

Mr. Aya-ay: Oo. Kasi minsan sinasabi lang namin maghintay sa labas, pagtingin naman namin, wala, mga umaalis tapos maya-maya babalik. Syempre, yung priority namin kung sino yung nandiyan, kasi wala namang proper reservation talaga.

Morcozo: Pwede ba magpa-reserve during peak hours?

Mr. Aya-ay: Pwede naman sila magpa-reserve.

Morcozo: Paano niyo po bina-balance yung reservation niyo sa walk-in, mas inuuna niyo po ba yung reservation niyo kaysa sa walk-in?

Mr. Aya-ay: Kung nagpa-reserve sila, then after 15 minutes wala pa sila sa binigay nilang oras, ibibigay na namin yung table sa mga walk-in.

Morcozo: Specific table po ba iyon or next available table lang?

Mr. Aya-ay: Depende kung may gusto silang table or kung sinabi nila na kahit saan nalang.

Morcozo: So sa reservation, pwede sila mamili ng table or kahit anong available table?

Mr. Aya-ay: Oo, pwede sila mamili o kahit anong table.

Morcozo: Ano po yung yung problem na nae-encounter niyo sa set-up ng reservation niyo ngayon?

Mr. Aya-ay: Ang hassle dun is yung nagsu-sunod sunod sila. Kunyari parehas ng time ng reservation then parehas din ng table, mahihirapan kami nun. Kailangan pa namin malaman kung saang table pa available.

Morcozo: Nakaka-experience po ba kayo ng double bookings or nami-miss na messages kasi hindi nache-check ng maayos?

Mr. Aya-ay: Oo, minsan ganun. Yung iba kasi nagcha-chat sa page, e wala pa akong access sa page. So need pa akong i-message ng may-ari para lang malaman ko na may nagpa-reserve.

Morcozo: If may website na po, itutuloy niyo pa rin po ba yung reservation sa facebook or sa buong website nalang?

Mr. Aya-ay: Mas maganda kung sa website nalang para centralized na and automatic.

Morcozo: Magkano po yung per hour ng table niyo?

Mr. Aya-ay: Sa regular, 150. Sa VIP naman, 200 siya. Pero depende kung regular customer namin, pwede namin gawing 150 ang VIP kung lagi naman silang naglalaro dito.

Morcozo: Paano niyo po sila sinisingil?

Mr. Aya-ay: After ng game nila, tsaka namin sisingilin.

Morcozo: Pag naglaro po sila ng 1.5 hours, ang charge po ba noon ay pang 1.5 hour na?

Mr. Aya-ay: Kung exact 1.5 hours automatic na iyon ang price namin pang 1.5 hours na. Pero kapag lumagpas sila ng mga 10 minutes, 15 minutes, considered na iyon as 2 hours. Parehas na rin iyon kapag 1 hour ang laro nila.

Morcozo: May iba pa po ba kayong rates?

Mr. Aya-ay: Regular rate lang kami ngayon kasi tapos na yung april. Kasi nung april 100 kami per hour.

Morcozo: So parang meron po kayong mga promos?

Mr. Aya-ay: Oo.

Morcozo: Paano kayo nagt-track ng time ng mga table?

Mr. Aya-ay: May papel kami na binibigay na nakalagay doon yung time-in nila, tapos ibibigay na lang nila iyon pagkatapos ng laro nila. Doon namin ito tine-track, and may timer din kami para if ever na magkamali, doon namin icha-charge.

Morcozo: Naga-alarm po ba yung timer niyo?

Mr. Aya-ay: Hindi, nagru-running lang siya, tuloy tuloy.

Morcozo: Paano po yung process ng papel?

Mr. Aya-ay: Nakalagay kasi doon yung date, time-in, time-out, name ng customer, and name ng staff. Ilalagay namin yung date and yung time-in then ibibigay sa customer, babalik nalang samin yung papel kapag tapos na sila, then tsaka namin ilalagay yung time-out. Doon namin tine-track kung nakaka-ilang hours sila and para mabayaran na nila.

Morcozo: Ganon din po ba yung process niyo sa reservations?

Mr. Aya-ay: Oo, ganun din. Kapag nandito na sila binibigyan din namin sila ng papel para ma-track yung oras nila.

Morcozo: Paano niyo ginagawa kapag madami ng tao, nagkakaroon po ba kayo ng problema sa tracking?

Mr. Aya-ay: Sa tracking, hindi namin sila ina-update sa time nila. Binibigyan nalang namin sila ng papel and nakalagay doon yung oras nila para ma-track din nila yung time. So kapag nakita na nilang time na sila, lalapit na sila sa akin para magbayad.

Morcozo: Paano po kapag time na nila pero hindi pa sila lumalapit, kayo na po ba yung lalapit?

Mr. Aya-ay: Hindi na ako lalapit, kumbaga ang dating nun kapag tapos na yung oras nila, is gusto pa nila maglaro, so tuloy-tuloy pa rin yung oras nila. Kasi sinasabi ko naman sakanila, tuloy-tuloy niyo lang oras niyo then kapag gusto na nila mag-out, tsaka na sila lalapit sa akin.

Morcozo: So bini-bill niyo po sila base sa oras na nagamit nila?

Mr. Aya-ay: Oo.

Morcozo: Kapag puno na po yung mga table and may-pumapasok pa na mga customer, ano po yung sinasabi niyo? Paano po sila naghi-hintay?

Mr. Aya-ay: Chine-check namin yung timer namin. May mga nagsasabi kasi ng 1 hour or 2 hours lang and then inaalala namin iyon then kapag nakita namin sa timer na 30 minutes nalang magiging available na itong mesa, tsaka namin sila sinasabihan. Pero normally ang ginagawa namin, kapag nakita namin 30 minutes pa magkakaroon ng available na table, sasabihin naming 40 minutes for extra time/waiting time.

Morcozo: Paano niyo po hina-handle yung pila kapag puno na kayo?

Mr. Aya-ay: Ginagawa namin, kunyari dalawa yung waiting, sinasabi namin “pangalawa po kayo sa waiting” and then sinasabi namin kung ilang minutes silang maghihintay para doon. Kasi may sabay-sabay namang mag-ooout dyan sa mga table.

Morcozo: Paano niyo po nakikilala yung mga customer na nasa pila?

Mr. Aya-ay: Yung iba naman kilala na namin since regular sila. Kapag hindi namin sila kilala, ililista namin yung pangalan sa papel and then tatawagin nalang namin.

Morcozo: May limit ba yung tao sa loob ng establishment?

Mr. Aya-ay: Wala naman, pwede kahit madami, basta hindi lang siksikan. Once na mapansin na namin sobrang sikip na, tsaka kami nagpapalabas ng mga hindi naglalaro, yung mga nakatambay lang.

Morcozo: Anong oras po kayo hindi na kumukuha ng reservation or walk-in?

Mr. Aya-ay: 11PM hindi na kami kumukuha. Yung cut off naman namin is 12PM, kapag nareach na yung 12PM, hinihintay nalang namin matapos yung mga tables then tsaka na kami maglilinis tapos out na.

Morcozo: So bawal na po talaga sila mag extend and bawal na mag walk-in?

Mr. Aya-ay: Oo.

Morcozo: Paano po kapag nagsabi yung magpapareserve na 30 minutes silang mala-late?

Mr. Aya-ay: Hindi pwede na 30 minutes silang mala-late. Gagawin namin kapag time na nila, itatanong namin kung uumpisahan na talaga yung oras nila, para wala ng maglalaro sa table. Kapag sinabi naman nila na hindi na sila matutuloy, ibibigay na namin sa walk-in yung oras nila.

Morcozo: Ilang minutes niyo po hinihintay yung reply nila?

Mr. Aya-ay: Mga 15 minutes lang. kapag wala ng reply, binibigay na namin yung table nila.

Morcozo: Paano po kung after 15 minutes bigla silang dumating?

Mr. Aya-ay: Late kasi sila, so io-offer nalang namin sila sa ibang table or sa next na magiging available na table.

Morcozo: So kung wala na pong available na tables after nilang dumating ng 15 minutes late, na di po sila nakapag reply at nabigay na po yung table nila sa ibang customer, ano pong mangyayari po sa kanila? Especially if may mahabang pila na po. Mauuna po ba sila sa pila or mahuhuli?

Mr. Aya-ay: Since late na sila, and na missed out lang naman nila yung reservation nila. Sila nalang yung isusunod namin sa next na magiging available na table.

Morcozo: Tungkol naman po sa cancellation, anong oras po yung pinaka late na pwede silang mag cancel ng reservation? And also for rescheduling?

Mr. Aya-ay: 1 hour before their reservation time ang pag-cancel para hindi masyado maapektuhan yung mga walk-ins namin na maglalaro at naglalaro sa isang araw.

Morcozo: Kapag naman po nagtawag kayo ng walk-in customer tapos walang pakita, ano pong ginagawa niyo doon?

Mr. Aya-ay: Kapag wala siya automatic binibigay na namin sa ibang waiting customer. Kapag naman bumalik sila, nandoon pa rin yung pila nila, so sila pa rin yung next na maglalaro pag nagkaroon ng available table.

Morcozo: May time limit po ba para bumalik sila?

Mr. Aya-ay: Same lang sa reservation policy namin, 15 minutes lang din.

Morcozo: Paano naman po kapag tapos na yung oras ng isang table, tapos gusto pa nilang mag-extend, tapos dumating na yung naka-reserve sa table, ano pong ginagawa niyo doon?

Mr. Aya-ay: Hindi na namin sila pinapayagan kasi dumating na yung naka-reserve. Kapag madaming tao sa pila, sinusunod namin yung oras nila para ma-accomodate namin yung ibang tao sa pila.

Morcozo: Paano po kapag punuan na tapos may naka-reserve sa table na ito sa next 1 hour, tapos may dumating na gusto magpa-reserve ng 2 hours, ano pong ginagawa niyo para ma-ensure na available yung table sa naka-reserve.

Mr. Aya-ay: Sinasabi namin sa customer na “1 hour nalang po yung available dyan since may naka-reserve na po sa following hours”. Para ma-sigurado namin na available itong table na to sa naka-reserve. Kung gusto naman nila mag 2 hours, ang ginagawa namin is ino-offer nalang namin sila sa magt-time na table.

Morcozo: Pag nakasira ng billiard equipment yung customer ano pong ginagawa niyo?

Mr. Aya-ay: Pinapa bayaran namin sakanila. Need nila bayaran kasi mahal yung mga gamit ng billiards.

Morcozo: May internet po ba dito sa One Shot?

Mr. Aya-ay: Meron.

Morcozo: Free po ba yung wifi sa lahat, pati customer pwede?

Mr. Aya-ay: Hindi, para lang dito sa mga tao sa One Shot.

Morcozo: Stable naman po ba yung internet connection?

Mr. Aya-ay: Stable naman. Nawala lang kahapon.

Morcozo: So may tendencies na mawalan ng internet connection. Noted. Sino po yung mga nago-operate ng system niyo?

Mr. Aya-ay: Ako lang, ako lang naman nag-hahandle masyado dyan sa loob.

Morcozo: Meron po ba kayong analytics sa system niyo ngayon?

Mr. Aya-ay: Lumalabas dun yung nakita naming pera kasama yung cash and gcash and then ico-compute nalang namin yon para makuha yung overall total na nakita sa araw iyon.

Morcozo: Pwede po ba mang hiram ng mga tako yung customer?

Mr. Aya-ay: Oo. May ino-offer kaming mga tako na pang house key namin.

Morcozo: Aside po sa default na tig-iisa silang tako per player, ilan po yung maximum nila na pwede magamit?

Mr. Aya-ay: Sa isang table kasi, dalawang tako lang yung kaya namin i-offer.

Morcozo: Yung paglilipat ng table, pwede yun kapag available lang?

Mr. Aya-ay: Oo. Kapag available na yung table na iyon, pwede lumipat dun yung mga customer kung gusto nila.

Morcozo: Yun lang naman po, Thank you po.

Mr. Aya-ay: Thank you.

CERTIFICATION OF TRANSCRIPT ACCURACY

I hereby certify that I have read the foregoing transcript of the interview, and I acknowledge that it is a true, accurate, and complete record of my statements and answers. I grant the research proponents permission to use the information provided herein for the sole purpose of their Capstone Project requirement.

Conforme:

A handwritten signature in black ink, appearing to read 'Eissen Justine Aya-Ay', written over the printed name.

MR. EISSEN JUSTINE AYA-AY

Interviewee

APPENDIX B. PERSONAL TECHNICAL VITAE

Curriculum Vitae of
Zeji S. Arnoco
Block 23 Suha Street, Barangay San Juan, Taytay, Rizal
zejiarnoco12@gmail.com
09604830893

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept 2023-Present	STI College Ortigas-Cainta
Vocational/Technical	2021-2023	STI College Ortigas-Cainta
High School	2017-2021	Christ the King College of Angono
Elementary	2011-2017	Christ the King College of Angono

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2025-Present	ALPHA (Alliance of Leading Programmers through Heuristic Adaptation)	Assistant Secretary of Logistics

SKILLS

SKILLS	Level of Competency	Date Acquired
N/A	N/A	N/A

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
N/A	N/A

Curriculum Vitae of
Karl Andrei S. De Guzman
039 Dona Aurora Street, Barangay San Roque, Angono, Rizal
dreisison420@gmail.com
09692395228

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept 2023-Present	STI College Ortigas-Cainta
Vocational/Technical	2021-2023	Laguna College of Business and Arts
High School	2017-2021	Laguna College of Business and Arts
Elementary	2011-2017	Meldenfield Christian Academy

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

SKILLS	Level of Competency	Date Acquired
Java	7/10	2021
Python	4/10	2023
SSMS	8/10	2024
HTML	9/10	2019

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
N/A	N/A

Curriculum Vitae of
Michael H. Grace
43 Philip St. Vista Verde Executive Village, Cainta, Rizal
mhgracek09@gmail.com
09565765167

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept 2023-Present	STI College Ortigas-Cainta
Vocational/Technical	2021-2023	San Lorenzo Ruiz Senior High School
High School	2017-2021	Manggahan High School
Elementary	2011-2017	Sto. Niño Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2025- Present	ALPHA (Alliance of Leading Programmers through Heuristic Adaptation)	Logistics Associate

SKILLS

SKILLS	Level of Competency	Date Acquired
N/A	N/A	N/A

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
2022-2023	TESDA NCII - Computer Systems Servicing

Curriculum Vitae of
Mark Joseph D. Morcozo
BLK 8 LT 7, Texas St. Duraville Homes Phase II, Mahabang Parang, Angono, Rizal
Morcozo.332694@ortigas-cainta.sti.edu.ph
09938326865

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept 2023-Present	STI College Ortigas-Cainta
Vocational/Technical	2021-2023	San Jose National High School
High School	2016-2021	Golden Sunbeams Christian School
Elementary	2015-2016	Golden Sunbeams Christian School
Elementary	2012-2015	Dalig Elementary School
Elementary	2010-2012	Doña Nieves Memorial Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
September 2025- Present	Audio Ingress/Egress	Minsan Studio

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
September 2025- Present	STI College Student Government	Events Committee Head
September 2024- Present	Aperture Media	Photographer/Editor
October 2023- 2025	ALPHA (Alliance of Leading Programmers through Heuristic Adaptation)	Secretary of Multimedia

SKILLS

SKILLS	Level of Competency	Date Acquired
Web Development	7/10	Sept 2026
Database Management	6/10	August 2025
Event-Driven	8/10	August 2024
OOP	8/10	August 2024

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
June 2025-2026	TESDA TWSP NCII – Computer Systems Servicing